Print sheet

Generated 2026-11-04 13:39. Bed 256 x 256 x 260 mm (Bambu X2D). Masses and times are rough estimates: walls and infill per material from the settings below, volume at a typical print rate plus 5 s per 0.2 mm layer, the layer time counted once per plate.

Warnings

launch_guard_l_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

launch_guard_r_a: print bbox 250.0 x 49.0 fits the bed but not inside the 5 mm margin; printed alone, centred on its own plate

Parts

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
frame_deck_2	PLA	1	222.4 x 211.8 x 8.0	360.4	304.3	7.63	0	no
frame_deck_4	PLA	1	214.4 x 211.8 x 8.0	358.9	301.6	7.56	0	no
frame_deck_1	PLA	1	222.4 x 168.8 x 8.0	276.2	233.6	5.87	0	no
frame_deck_3	PLA	1	214.4 x 168.8 x 8.0	274.2	231.4	5.81	0	no
frame_cheek_r	PLA	1	215.0 x 220.0 x 30.0	248.2	205.2	5.32	1084	yes, tree supports from the plate (1084 mm2 over 45 deg)
frame_camera_mount	PLA	1	58.0 x 72.0 x 236.4	236.8	166.5	5.79	672	yes, tree supports from the plate (672 mm2 over 45 deg)
launch_aim_housing	PLA	1	176.5 x 59.0 x 183.0	212.8	185.4	5.89	761	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	92.5 x 68.0 x 170.0	201.0	178.7	5.63	2356	yes, tree supports from the plate (2356 mm2 over 45 deg)
frame_motor_leg	PLA	1	49.0 x 69.0 x 187.0	159.9	117.3	4.22	222	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_front	PLA	1	131.5 x 128.0 x 48.5	146.5	137.7	3.76	7890	yes, tree supports from the plate (7890 mm2 over 45 deg)
frame_ebay	PLA	1	124.0 x 124.0 x 58.0	141.3	175.3	4.77	879	yes, tree supports from the plate (879 mm2 over 45 deg)
drum_drum_a	PLA	1	97.1 x 97.1 x 43.0	138.6	103.2	2.87	719	yes, tree supports from the plate (719 mm2 over 45 deg)
launch_rail_rear	PLA	1	151.2 x 68.0 x 40.1	132.0	115.9	3.16	3676	yes, tree supports from the plate (3676 mm2 over 45 deg)
reel_bridge	PLA	1	220.0 x 192.1 x 88.0	114.1	118.5	3.56	686	yes, tree supports from the plate (686 mm2 over 45 deg)
reel_head_front	PLA	1	188.0 x 186.0 x 6.0	109.3	108.4	2.74	50	no

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
reel_head_spine	PLA	1	130.8 x 188.0 x 6.0	106.9	104.2	2.64	0	no
drum_drum_b	PLA	1	76.7 x 76.7 x 41.0	105.9	82.4	2.34	3126	yes, tree supports from the plate (3126 mm2 over 45 deg)
reel_head_p2	PLA	1	104.3 x 182.0 x 6.0	96.0	94.1	2.38	20	no
launch_stop_spring_l	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
launch_stop_spring_r	PLA	1	154.9 x 51.5 x 14.0	65.8	81.6	2.13	57	no
frame_ebay_lid	PLA	1	124.0 x 124.0 x 7.0	57.6	71.4	1.83	80	no
reel_head_p1	PLA	1	100.0 x 177.0 x 37.0	55.4	55.0	1.63	98	no
launch_guard_l_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_guard_r_a	PLA	1	250.0 x 49.0 x 40.0	52.4	65.0	1.90	93	no
launch_carriage	PLA	1	205.2 x 240.2 x 43.0	48.0	59.6	1.78	6453	yes, tree supports from the plate (6453 mm2 over 45 deg)
frame_guard_front	PLA	1	155.0 x 104.0 x 16.4	45.8	56.8	1.53	40	no
drum_index_block	PLA	1	25.0 x 68.0 x 39.5	41.0	33.2	1.10	61	no
frame_can_stay	PLA	1	55.0 x 55.0 x 71.0	31.0	28.4	1.20	906	yes, tree supports from the plate (906 mm2 over 45 deg)
reel_spool	PLA	1	62.0 x 62.0 x 22.4	28.4	30.8	0.92	1791	yes, tree supports from the plate (1791 mm2 over 45 deg)
launch_stop_frame_l	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	174.6 x 53.0 x 55.1	26.3	32.7	1.20	541	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	76.1 x 108.6 x 6.0	24.4	25.6	0.68	24	no
drum_switch_block	PLA	1	26.0 x 66.0 x 21.0	20.1	18.3	0.60	216	yes, tree supports from the plate (216 mm2 over 45 deg)
launch_carriage_yoke	PLA	1	43.3 x 126.6 x 28.0	18.9	23.5	0.78	1406	yes, tree supports from the plate (1406 mm2 over 45 deg)
frame_rc832_holder	PLA	1	97.0 x 21.6 x 40.0	18.4	22.8	0.84	0	no
reel_head_mid	PLA	1	53.8 x 58.5 x 6.0	18.1	19.3	0.52	143	yes, tree supports from the plate (143 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145.3 x 32.5 x 40.0	17.6	21.8	0.82	56	no
drum_fork	PLA	1	86.0 x 68.5 x 51.0	16.6	19.9	0.85	1227	yes, tree supports from the plate (1227 mm2 over 45 deg)

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
sensors_sw_rail_low_bracket	PLA	1	31.0 x 48.0 x 74.0	16.1	17.8	0.96	65	no
launch_guard_l_b2	PLA	1	145.3 x 32.5 x 40.0	14.4	17.9	0.72	33	no
sensors_as5600_carrier	PLA	1	34.0 x 67.5 x 11.0	12.4	12.7	0.39	90	no
frame_guard_hood	PLA	1	122.0 x 56.1 x 14.4	11.4	14.1	0.45	84	no
reel_cup	PLA	1	48.0 x 53.4 x 18.0	10.2	12.7	0.44	1210	yes, tree supports from the plate (1210 mm2 over 45 deg)
drum_n20_cradle	PLA	1	53.5 x 32.0 x 19.5	9.7	12.1	0.44	114	yes, tree supports from the plate (114 mm2 over 45 deg)
reel_head_p1b	PLA	1	38.0 x 44.5 x 16.0	9.0	10.7	0.38	41	no
sensors_sw_gap	PLA	1	63.5 x 43.3 x 11.5	8.0	9.9	0.33	64	no
reel_shroud	PLA	1	70.0 x 65.9 x 29.5	6.1	7.6	0.39	0	no
sensors_sw_rail_low	PLA	1	48.0 x 35.0 x 11.5	5.9	7.3	0.26	22	no
drum_dog_post	PLA	1	21.8 x 65.0 x 48.2	5.7	7.1	0.51	90	no
reel_sw_base	PLA	1	46.0 x 35.0 x 11.6	5.7	7.0	0.26	22	no
drum_follower	PLA	1	25.7 x 38.6 x 17.0	5.2	6.2	0.27	260	yes, tree supports from the plate (260 mm2 over 45 deg)
launch_aim_cap	PLA	1	45.5 x 14.0 x 19.4	4.2	5.2	0.27	0	no
launch_guard_l_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_r_c	PLA	1	32.0 x 54.5 x 26.0	3.8	4.8	0.30	3	no
launch_guard_l_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
launch_guard_r_b1	PLA	1	11.7 x 49.0 x 40.0	3.6	4.5	0.39	22	no
frame_cable_anchor_1	PLA	1	16.0 x 24.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
frame_cable_anchor_2	PLA	1	24.0 x 16.0 x 10.0	3.2	4.0	0.17	141	yes, tree supports from the plate (141 mm2 over 45 deg)
drum_link_eye_1	PLA	1	13.0 x 13.0 x 23.0	2.5	3.0	0.24	29	no
drum_link_eye_2	PLA	1	13.0 x 13.0 x 23.0	2.5	3.0	0.24	29	no
drum_n20_strap	PLA	1	53.5 x 17.9 x 3.0	2.2	2.7	0.09	0	no
drum_magnet_cap	PLA	1	20.0 x 20.0 x 6.5	1.5	1.9	0.09	40	no
frame_ebay_esp_clamp	PLA	1	43.4 x 8.0 x 4.0	1.3	1.6	0.07	0	no
sensors_sw_gap_lever	PLA	1	40.0 x 9.0 x 8.0	1.1	1.4	0.09	32	no
sensors_sw_rail_low_lever	PLA	1	44.0 x 5.0 x 8.0	1.1	1.3	0.09	32	no
reel_sw_lever	PLA	1	42.0 x 5.0 x 8.0	1.0	1.3	0.09	32	no
sensors_as5600_strip_top	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
sensors_as5600_strip_bot	PLA	1	25.4 x 11.8 x 3.1	0.8	1.0	0.05	0	no
drum_index_lever	PLA	1	12.9 x 47.5 x 6.0	0.7	0.9	0.06	94	no
drum_sw_lever_0	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
drum_sw_lever_1	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no
drum_sw_lever_2	PLA	1	13.7 x 47.5 x 6.0	0.7	0.9	0.06	97	no
launch_sw_seated	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_sw_home	PLA	1	41.5 x 8.4 x 6.0	0.7	0.8	0.06	24	no
launch_muzzle_roller	PLA	1	14.0 x 14.0 x 5.0	0.4	0.5	0.05	64	no
launch_gear_washer	PLA	1	18.0 x 18.0 x 2.3	0.4	0.5	0.03	0	no
launch_aim_drag_pad	PLA	1	8.8 x 24.5 x 4.0	0.3	0.4	0.04	0	no
launch_gear_clip	PLA	1	17.3 x 18.0 x 1.4	0.2	0.3	0.02	0	no
launch_rail_splice_l	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_rail_splice_r	PLA	1	22.0 x 5.0 x 3.0	0.2	0.3	0.03	0	no
launch_drive_washer	PLA	1	32.0 x 32.0 x 0.7	0.2	0.2	0.01	0	no
drum_shear_pin	PLA	6	12.1 x 5.0 x 4.1	0.1	0.2	0.03	0	no
reel_cup_clip	PLA	1	10.0 x 9.7 x 2.4	0.1	0.2	0.02	0	no
launch_aim_drag_shim	PLA	1	8.6 x 7.0 x 4.0	0.1	0.1	0.03	0	no
launch_aim_idler_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_aim_crankpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_aim_lugpin_clip	PLA	1	10.8 x 10.0 x 1.4	0.1	0.1	0.01	0	no
launch_seat_clip	PLA	1	5.6 x 5.6 x 2.4	0.1	0.1	0.02	0	no
launch_muzzle_washer_l	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
launch_muzzle_washer_r	PLA	1	6.2 x 6.2 x 0.9	0.0	0.0	0.01	0	no
frame_motor_adapter	PPA-CF	1	194.0 x 194.0 x 16.0	282.2	302.8	13.57	5484	yes, tree supports from the plate (5484 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105.0 x 105.0 x 27.3	184.2	177.6	8.09	639	yes, tree supports from the plate (639 mm2 over 45 deg)
drum_gear_reel	PPA-CF	1	104.0 x 103.9 x 19.0	87.9	95.9	4.39	0	no
reel_c1	PPA-CF	1	101.9 x 102.0 x 12.0	87.4	93.7	4.25	0	no
drum_coupler	PPA-CF	1	78.0 x 77.9 x 61.5	79.6	89.3	4.40	1998	yes, tree supports from the plate (1998 mm2 over 45 deg)
drum_shaft	PPA-CF	1	170.3 x 25.4 x 22.0	42.7	45.2	2.16	922	yes, tree supports from the plate (922 mm2 over 45 deg)
reel_drum	PPA-CF	1	46.0 x 46.0 x 43.0	41.5	45.0	2.30	251	yes, tree supports from the plate (251 mm2 over 45 deg)
launch_sleeve	PPA-CF	1	64.0 x 64.0 x 85.0	39.4	49.3	2.78	2907	yes, tree supports from the plate (2907 mm2 over 45 deg)
drum_collar	PPA-CF	1	60.0 x 60.0 x 26.0	38.5	44.0	2.14	2694	yes, tree supports from the plate (2694 mm2 over 45 deg)

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
launch_aim_crank	PPA-CF	1	62.3 x 59.6 x 38.0	28.9	35.3	1.83	985	yes, tree supports from the plate (985 mm2 over 45 deg)
drum_idler_gear	PPA-CF	1	76.5 x 76.5 x 7.0	28.7	35.8	1.64	0	no
launch_drive_gear	PPA-CF	1	70.0 x 69.7 x 28.8	23.9	29.9	1.53	2457	yes, tree supports from the plate (2457 mm2 over 45 deg)
reel_c1p	PPA-CF	1	72.0 x 71.9 x 9.6	22.3	27.8	1.30	0	no
launch_aim_output	PPA-CF	1	48.0 x 47.8 x 59.5	21.8	26.2	1.58	0	no
reel_uc	PPA-CF	1	10.0 x 10.0 x 243.5	19.0	23.7	2.75	4	no
launch_aim_worm	PPA-CF	1	28.0 x 28.0 x 54.5	17.1	21.3	1.33	2779	yes, tree supports from the plate (2779 mm2 over 45 deg)
drum_dogring_launch	PPA-CF	1	62.0 x 62.0 x 10.5	15.2	19.0	0.92	123	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_sun	PPA-CF	1	48.0 x 47.8 x 12.0	14.6	18.3	0.89	0	no
reel_c	PPA-CF	1	22.6 x 20.0 x 58.5	13.9	17.4	1.18	33	no
launch_aim_lug	PPA-CF	1	30.0 x 58.0 x 27.8	13.6	16.9	0.95	995	yes, tree supports from the plate (995 mm2 over 45 deg)
reel_gb	PPA-CF	1	51.0 x 51.0 x 7.0	12.1	15.1	0.72	0	no
reel_gk	PPA-CF	1	45.0 x 45.0 x 12.4	12.0	15.0	0.75	1031	yes, tree supports from the plate (1031 mm2 over 45 deg)
drum_coupler_hub	PPA-CF	1	26.0 x 26.0 x 32.5	11.7	14.6	0.88	37	no
launch_aim_conrod	PPA-CF	1	173.0 x 17.0 x 6.0	11.3	14.2	0.67	0	no
reel_cs_c	PPA-CF	1	22.6 x 20.0 x 40.5	10.2	11.4	0.79	182	yes, tree supports from the plate (182 mm2 over 45 deg)
reel_sprag_hub	PPA-CF	1	56.0 x 56.0 x 14.0	9.8	12.2	0.64	33	no
reel_gi	PPA-CF	1	48.0 x 47.8 x 6.0	9.1	11.4	0.55	0	no
reel_k	PPA-CF	1	10.0 x 10.0 x 108.6	8.1	10.2	1.21	14	no
reel_ga	PPA-CF	1	42.0 x 41.8 x 7.0	7.7	9.6	0.47	0	no
drum_axle	PPA-CF	1	95.3 x 10.0 x 9.0	7.1	8.9	0.46	0	no
drum_rod	PPA-CF	1	95.3 x 10.0 x 9.0	7.1	8.9	0.46	0	no
launch_aim_bevel30	PPA-CF	1	52.6 x 52.4 x 9.7	7.0	8.8	0.46	590	yes, tree supports from the plate (590 mm2 over 45 deg)
drum_idler	PPA-CF	1	22.0 x 18.6 x 54.5	7.0	8.7	0.77	331	yes, tree supports from the plate (331 mm2 over 45 deg)
drum_shift_rail	PPA-CF	1	88.0 x 10.0 x 9.0	6.5	8.2	0.43	0	no
reel_d	PPA-CF	1	53.0 x 53.0 x 3.0	6.0	7.6	0.36	1050	yes, tree supports from the plate (1050 mm2 over 45 deg)
launch_aim_idler	PPA-CF	1	30.0 x 29.7 x 12.0	6.0	7.5	0.42	0	no
drum_crank	PPA-CF	1	56.5 x 57.5 x 9.6	5.5	6.9	0.37	0	no

part	material	qty	print bbox mm	vol cm3	mass g	time h	overhang mm2	supports
reel_lever	PPA-CF	1	67.0 x 66.0 x 8.0	5.5	6.9	0.36	20	no
launch_aim_arm	PPA-CF	1	63.6 x 41.0 x 19.3	5.3	6.6	0.43	14	no
reel_ut	PPA-CF	1	8.0 x 8.0 x 104.5	5.2	6.5	1.01	4	no
reel_s	PPA-CF	1	99.0 x 8.0 x 7.0	4.5	5.7	0.30	0	no
launch_gear_pin	PPA-CF	1	53.0 x 16.0 x 12.5	4.3	5.4	0.33	68	no
reel_gc	PPA-CF	1	37.4 x 37.5 x 7.0	4.3	5.3	0.29	0	no
reel_t1	PPA-CF	1	33.0 x 33.0 x 6.0	3.7	4.6	0.25	0	no
reel_swing_arm	PPA-CF	1	72.6 x 19.0 x 9.6	2.5	3.2	0.21	0	no
reel_cs_t	PPA-CF	1	12.0 x 12.0 x 24.5	2.2	2.8	0.29	39	no
reel_b1g	PPA-CF	1	39.0 x 39.0 x 6.2	2.1	2.6	0.16	679	yes, tree supports from the plate (679 mm2 over 45 deg)
reel_b3b	PPA-CF	1	22.0 x 22.0 x 8.8	1.1	1.4	0.12	0	no
reel_b2p	PPA-CF	1	22.0 x 22.0 x 7.8	1.0	1.3	0.11	149	yes, tree supports from the plate (149 mm2 over 45 deg)
reel_b3a	PPA-CF	1	22.0 x 22.0 x 7.8	1.0	1.3	0.11	0	no
launch_yoke_peg_l	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
launch_yoke_peg_r	PPA-CF	1	20.0 x 8.0 x 7.4	1.0	1.2	0.11	0	no
reel_b1p	PPA-CF	1	21.0 x 21.0 x 11.4	1.0	1.2	0.13	87	no
reel_stub	PPA-CF	1	20.0 x 8.0 x 6.8	0.9	1.1	0.10	0	no
reel_t	PPA-CF	1	18.0 x 18.0 x 5.6	0.9	1.1	0.09	0	no
launch_seat_plunger	PPA-CF	1	29.5 x 15.0 x 19.6	0.8	1.0	0.18	0	no
drum_reel_washer	PPA-CF	1	35.1 x 35.1 x 1.4	0.8	0.9	0.05	0	no
reel_b2g	PPA-CF	1	22.0 x 22.0 x 6.8	0.7	0.9	0.09	149	yes, tree supports from the plate (149 mm2 over 45 deg)
drum_n20_pinion	PPA-CF	1	9.8 x 9.5 x 8.0	0.4	0.5	0.08	0	no
launch_seat_pushrod	PPA-CF	1	16.0 x 4.8 x 3.6	0.1	0.2	0.03	0	no

frame_deck_2: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 12 x M3 rover holes (it is its own drill template)
frame_deck_4: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 14 x M3 rover holes (it is its own drill template)
frame_deck_1: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template)
frame_deck_3: As modelled, bottom face on the bed. deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_cheek_r: Rotated: the part's -y axis points up. motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
frame_camera_mount: As modelled, bottom face on the bed. Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
launch_aim_housing: As modelled, bottom face on the bed. Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_rail_mid: Rotated: the part's -x axis points up. Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
frame_motor_leg: As modelled, bottom face on the bed. motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
launch_rail_front: As modelled, bottom face on the bed. Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
frame_ebay: As modelled, bottom face on the bed. electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors

drum_drum_a: As modelled, bottom face on the bed. Drum A: N20 gear 120T m0.8, groove 1 barrel (follower), hex spigot
launch_rail_rear: Rotated: the part's -z axis points up. Rail rear piece: shoe track, walls, flange at the rear end for the reel head (D49)
reel_bridge: Rotated: the part's +y axis points up. Rail bridge: plate inboard of the main-axis stack with countershafts C and T, the c1' stub, pillars to the u-shaft bushings, cad-drum's shift rail; box arm bolted to rail cheek A. Print on its inboard face
reel_head_front: Rotated: the part's -x axis points up. Head front plate: bolts the reel head to the rail rear flange, u-shaft T rear bushing, link window. Print flat
reel_head_spine: Rotated: the part's -y axis points up. Head spine: inboard wall from the front plate to P2; the drum brackets and plates screw to it. Print flat
drum_drum_b: Rotated: the part's -z axis points up. Drum B: brake cam (spool clutch lever), Gray code bands A B C, wrap segment gear + locking arc
reel_head_p2: Rotated: the part's -x axis points up. Head plate P2: cup shaft and wrap idler rear journals. Print flat
launch_stop_spring_l: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r: Rotated: the part's -z axis points up. Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
frame_ebay_lid: Rotated: the part's -z axis points up. bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
reel_head_p1: Rotated: the part's +x axis points up. Head plate P1: capstan tube, u-shaft C and G_b bushings, clutch lever post, spool shaft leg. Print flat
launch_guard_l_a: Rotated: the part's -y axis points up. Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_a: Rotated: the part's +y axis points up. Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_carriage: Rotated: the part's +y axis points up. Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
frame_guard_front: Rotated: the part's -x axis points up. front gear guard between the drum gears and the bay, M3 x 10 into the deck
drum_index_block: As modelled, bottom face on the bed. SW_INDEX block: column on the motor plate front face (2 x M3 x 10 into the frame inserts), M3 nut contact
frame_can_stay: Rotated: the part's -y axis points up. can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
reel_spool: Rotated: the part's -z axis points up. Spool d40 core, d62 flanges, 20 wide: 20 m of 1 mm Dyneema, turns free on two 608s. Print flange down
launch_stop_frame_l: Rotated: the part's -y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r: Rotated: the part's +y axis points up. Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
drum_bracket_b: As modelled, bottom face on the bed. Drum bracket B: axle, guide rod, wrap idler journal; foot screwed to the reel head spine
drum_switch_block: As modelled, bottom face on the bed. Code switch block on top of the drum: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)
launch_carriage_yoke: As modelled, bottom face on the bed. Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
frame_rc832_holder: As modelled, bottom face on the bed. RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
reel_head_mid: Rotated: the part's +y axis points up. Head mid plate: spool shaft inboard bushing. Print flat
launch_guard_r_b2: Rotated: the part's +y axis points up. Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
drum_fork: As modelled, bottom face on the bed. Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), lug for the bell crank pin, contact tab
sensors_sw_rail_low_bracket: As modelled, bottom face on the bed. SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
launch_guard_l_b2: Rotated: the part's -y axis points up. Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
sensors_as5600_carrier: Rotated: the part's -y axis points up. AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine
frame_guard_hood: Rotated: the part's -z axis points up. gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
reel_cup: As modelled, bottom face on the bed. Wrap cup (Harry v7, D49 axial eyelet at r 26): the line runs straight through the eyelet. Print open end down
drum_n20_cradle: As modelled, bottom face on the bed. N20 cradle on the reel head front plate: pivot + swing screw (hand fallback), SW_N20 contact
reel_head_p1b: Rotated: the part's +x axis points up. Head plate P1b: capstan tube rear bushing, SW_ARM. Print flat

sensors_sw_gap: As modelled, bottom face on the bed. SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
reel_shroud: Rotated: the part's +y axis points up. Spool shroud: keeps loose loops on the spool, open at the top front where the line leaves
sensors_sw_rail_low: As modelled, bottom face on the bed. SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point
drum_dog_post: Rotated: the part's -z axis points up. Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
reel_sw_base: As modelled, bottom face on the bed. SW_ARM contact base on P2's front face
drum_follower: As modelled, bottom face on the bed. Follower slider: pin in groove 1, slides along the guide rod; the link takes its motion to the bridge
launch_aim_cap: Rotated: the part's -z axis points up. Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_guard_l_c: Rotated: the part's -y axis points up. Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_c: Rotated: the part's +y axis points up. Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_l_b1: Rotated: the part's -y axis points up. Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1: Rotated: the part's +y axis points up. Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
frame_cable_anchor_1: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_cable_anchor_2: As modelled, bottom face on the bed. cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
drum_link_eye_1: Rotated: the part's -z axis points up. Link eye pressed on the rod end, M3 screw through the follower lug
drum_link_eye_2: Rotated: the part's +z axis points up. Link eye pressed on the rod end, M3 screw through the bell crank input arm
drum_n20_strap: As modelled, bottom face on the bed. End plate closing the N20 cradle, 2 x M3 x 8
drum_magnet_cap: Rotated: the part's -z axis points up. AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat
frame_ebay_esp_clamp: As modelled, bottom face on the bed. ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
sensors_sw_gap_lever: Rotated: the part's +y axis points up. SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
sensors_sw_rail_low_lever: Rotated: the part's +y axis points up. SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
reel_sw_lever: Rotated: the part's +y axis points up. SW_ARM lever: rides the cup gear's raised track (contact closed), drops into its home notch (open)
sensors_as5600_strip_top: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_as5600_strip_bot: Rotated: the part's +y axis points up. AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
drum_index_lever: As modelled, bottom face on the bed. SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
drum_sw_lever_0: As modelled, bottom face on the bed. Code switch lever A: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_1: As modelled, bottom face on the bed. Code switch lever B: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_2: As modelled, bottom face on the bed. Code switch lever C: flexure, round nub, M3 x 6 contact screw
launch_sw_seated: Rotated: the part's +y axis points up. SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_sw_home: Rotated: the part's +y axis points up. SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
launch_muzzle_roller: As modelled, bottom face on the bed. Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside
launch_gear_washer: As modelled, bottom face on the bed. Spacer washer: big gear hub to the left cheek
launch_aim_drag_pad: As modelled, bottom face on the bed. Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
launch_gear_clip: As modelled, bottom face on the bed. C clip on the big gear pin groove
launch_rail_splice_l: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r: As modelled, bottom face on the bed. Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_drive_washer: As modelled, bottom face on the bed. Spacer washer: drive gear hub to the left cheek
drum_shear_pin: Rotated: the part's -z axis points up. Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
reel_cup_clip: As modelled, bottom face on the bed. Harry's v7 C-clip on the cup shaft neck in front of the cup
launch_aim_drag_shim: As modelled, bottom face on the bed. Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
launch_aim_idler_clip: As modelled, bottom face on the bed. Printed C clip on the arm pin, holds the idler
launch_aim_crankpin_clip: As modelled, bottom face on the bed. Printed C clip on the crank pin, inboard of the conrod
launch_aim_lugpin_clip: As modelled, bottom face on the bed. Printed C clip on the lug pin, outboard of the conrod
launch_seat_clip: Rotated: the part's -z axis points up. Head cap on the plunger pin (rests on the yoke block)
launch_muzzle_washer_l: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r: Rotated: the part's -z axis points up. Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot

frame_motor_adapter: Rotated: the part's -z axis points up. puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, 2 inserts for cad-drum's SW_INDEX block; reprint when the motor is measured
launch_big_gear: Rotated: the part's -z axis points up. Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin
drum_gear_reel: As modelled, bottom face on the bed. REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar; meshes C1 on the rail bridge (D49)
reel_c1: As modelled, bottom face on the bed. C1 m2 49T: meshes the REEL output gear 50T across the pivot
drum_coupler: As modelled, bottom face on the bed. Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
drum_shaft: Rotated: the part's +y axis points up. Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
reel_drum: As modelled, bottom face on the bed. Capstan drum dia 40 x 40 (Harry v7), keyed on the tube. Print flange down
launch_sleeve: Rotated: the part's -z axis points up. Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
drum_collar: As modelled, bottom face on the bed. Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
launch_aim_crank: As modelled, bottom face on the bed. Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
drum_idler_gear: As modelled, bottom face on the bed. Wrap idler output gear 49T m1.5 (D bore) to the cad-reel cup gear 28T: 7 cup turns
launch_drive_gear: As modelled, bottom face on the bed. Drive gear m2.5 26T, sprag #1 inside on an integral key rib
reel_c1p: As modelled, bottom face on the bed. c1' m1.5 46T, free on its stub: driven by the bus gear 50T, drives the swing idler
launch_aim_output: As modelled, bottom face on the bed. Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
reel_uc: As modelled, bottom face on the bed. u-shaft C (PPA-CF d10, 244 long): along the rail from the B1 bevel to G_a. Print standing (244 of the 260 mm height); an 8 mm steel rod with printed sleeves is the stiff fallback
launch_aim_worm: As modelled, bottom face on the bed. Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
drum_dogring_launch: Rotated: the part's -z axis points up. LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
launch_aim_sun: As modelled, bottom face on the bed. Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
reel_c: As modelled, bottom face on the bed. Capstan tube C (PPA-CF d20): key rib in the drum, G_c on its flat, journals in P1 and P1b. Print standing on its rear end
launch_aim_lug: Rotated: the part's +y axis points up. Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
reel_gb: As modelled, bottom face on the bed. G_b m1.5 32T idler on its stub in P1
reel_gk: As modelled, bottom face on the bed. Cup gear G_k m1.5 28T: driven by the drum's wrap idler 49T (7 cup turns per wrap); the raised track on its rear face holds SW_ARM closed, the notch at home lets it open. Print track face down
drum_coupler_hub: As modelled, bottom face on the bed. Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
launch_aim_conrod: Rotated: the part's +y axis points up. Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
reel_cs_c: Rotated: the part's -z axis points up. Countershaft C (PPA-CF): sprag #2, C1 and the B1 bevel. Print standing
reel_sprag_hub: Rotated: the part's -z axis points up. Sprag #2 hub on the bridge: holds the CSK20PP capstan backstop
reel_gi: As modelled, bottom face on the bed. Swing idler m1.5 30T
reel_k: As modelled, bottom face on the bed. Cup shaft K (PPA-CF d10): runs inside the tube, cup in front, G_k behind P1b, journal in P2. Print standing on its rear end
reel_ga: As modelled, bottom face on the bed. G_a m1.5 26T on the rear end of u-shaft C
drum_axle: Rotated: the part's -x axis points up. Fixed drum axle d10, pressed into the reel head front plate and bracket B
drum_rod: Rotated: the part's +x axis points up. Follower guide rod d10, pressed into the reel head front plate and bracket B
launch_aim_bevel30: As modelled, bottom face on the bed. Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
drum_idler: As modelled, bottom face on the bed. Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, journals in bracket B and P2
drum_shift_rail: Rotated: the part's +x axis points up. Shift rail d10 on the rail bridge (pressed into the bridge plate boss)
reel_d: As modelled, bottom face on the bed. Clutch disk D on the spool shaft flat: its friction ring presses the spool's outboard face. Print ring up
launch_aim_idler: As modelled, bottom face on the bed. Splitter idler m1.5 z18 face 12 on the arm pin
drum_crank: Rotated: the part's -y axis points up. Bell crank 1:1 on the bridge shelf: turns the link's motion along the rail into the fork's shift
reel_lever: Rotated: the part's -x axis points up. Clutch lever (PPA-CF): the drum's brake cam presses its upper arm; the lower arm is the clutch spring leaf, its FIRM screw squeezes the spool. Print flat
launch_aim_arm: Rotated: the part's -z axis points up. Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
reel_ut: As modelled, bottom face on the bed. u-shaft T (PPA-CF d8): along the rail from the B2 bevel to the B3 bevel. Print standing
reel_s: Rotated: the part's +x axis points up. Spool shaft S (PPA-CF d8): B3 bevel and the clutch disk on its flats, journals in the mid plate and the P1 leg. Print lying on its 1 mm flat
launch_gear_pin: Rotated: the part's -x axis points up. Snap pin d10 for the big gear, C clip; prints lying on its flat
reel_gc: As modelled, bottom face on the bed. Capstan gear G_c m1.5 23T on the tube flat
reel_t1: As modelled, bottom face on the bed. T1 m1.5 20T: take-up at 2.5 x wiper
reel_swing_arm: Rotated: the part's -z axis points up. Swing arm on c1's hub: carries the idler into mesh with T1 on forward wiper turns
reel_cs_t: Rotated: the part's -z axis points up. Countershaft T (PPA-CF): T1 and the B2 bevel

reel_b1g: As modelled, bottom face on the bed. Bevel B1 gear m1.5 24T (2:1 up to u-shaft C)
reel_b3b: As modelled, bottom face on the bed. Bevel B3 m1 20T on the spool shaft
reel_b2p: As modelled, bottom face on the bed. Bevel B2 m1 20T on the front end of u-shaft T
reel_b3a: As modelled, bottom face on the bed. Bevel B3 m1 20T on the rear end of u-shaft T
launch_yoke_peg_l: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
launch_yoke_peg_r: Rotated: the part's +x axis points up. Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_b1p: As modelled, bottom face on the bed. Bevel B1 pinion m1.5 12T on the front end of u-shaft C
reel_stub: Rotated: the part's -x axis points up. c1' stub axle (PPA-CF d8, pressed into the bridge)
reel_t: As modelled, bottom face on the bed. Thrust ring: the lever's FIRM screw pushes it onto the inboard 608 inner race (clutch squeeze)
launch_seat_plunger: As modelled, bottom face on the bed. SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
drum_reel_washer: As modelled, bottom face on the bed. REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
reel_b2g: As modelled, bottom face on the bed. Bevel B2 m1 20T on countershaft T
drum_n20_pinion: As modelled, bottom face on the bed. N20 pinion 10T m0.8, D bore pressed on the N20 shaft
launch_seat_pushrod: Rotated: the part's -z axis points up. SW_SEATED pushrod d3.2 with the stem slot

Plates

plate	material	objects	mass g	time h
plate_1_PLA.3mf	PLA	7	95	2.6
plate_2_PLA.3mf	PLA	5	281	7.3
plate_3_PLA.3mf	PLA	6	318	8.1
plate_4_PLA.3mf	PLA	6	314	7.9
plate_5_PLA.3mf	PLA	6	174	4.9
plate_6_PLA.3mf	PLA	16	365	9.4
plate_7_PLA.3mf	PLA	4	450	12.4
plate_8_PLA.3mf	PLA	9	345	9.9
plate_9_PLA.3mf	PLA	4	250	6.6
plate_10_PLA.3mf	PLA	6	434	12.1
plate_11_PLA.3mf	PLA	12	372	9.7
plate_12_PLA.3mf	PLA	11	401	11.6
plate_13_PLA.3mf	PLA	1	82	2.3
plate_14_PLA.3mf	PLA	1	65	1.9
plate_15_PLA.3mf	PLA	1	65	1.9
plate_16_PPA-CF.3mf	PPA-CF	13	401	18.1
plate_17_PPA-CF.3mf	PPA-CF	18	569	26.0
plate_18_PPA-CF.3mf	PPA-CF	22	369	18.1
plate_19_PPA-CF.3mf	PPA-CF	7	118	5.7

Totals on the plates

material	objects	plates	mass g	filament m	time h
PLA	95	15	4011	1345.0	108.6
PPA-CF	60	4	1456	484.4	67.9

Recommended settings

PLA

- 0.4 mm nozzle, 0.2 mm layers

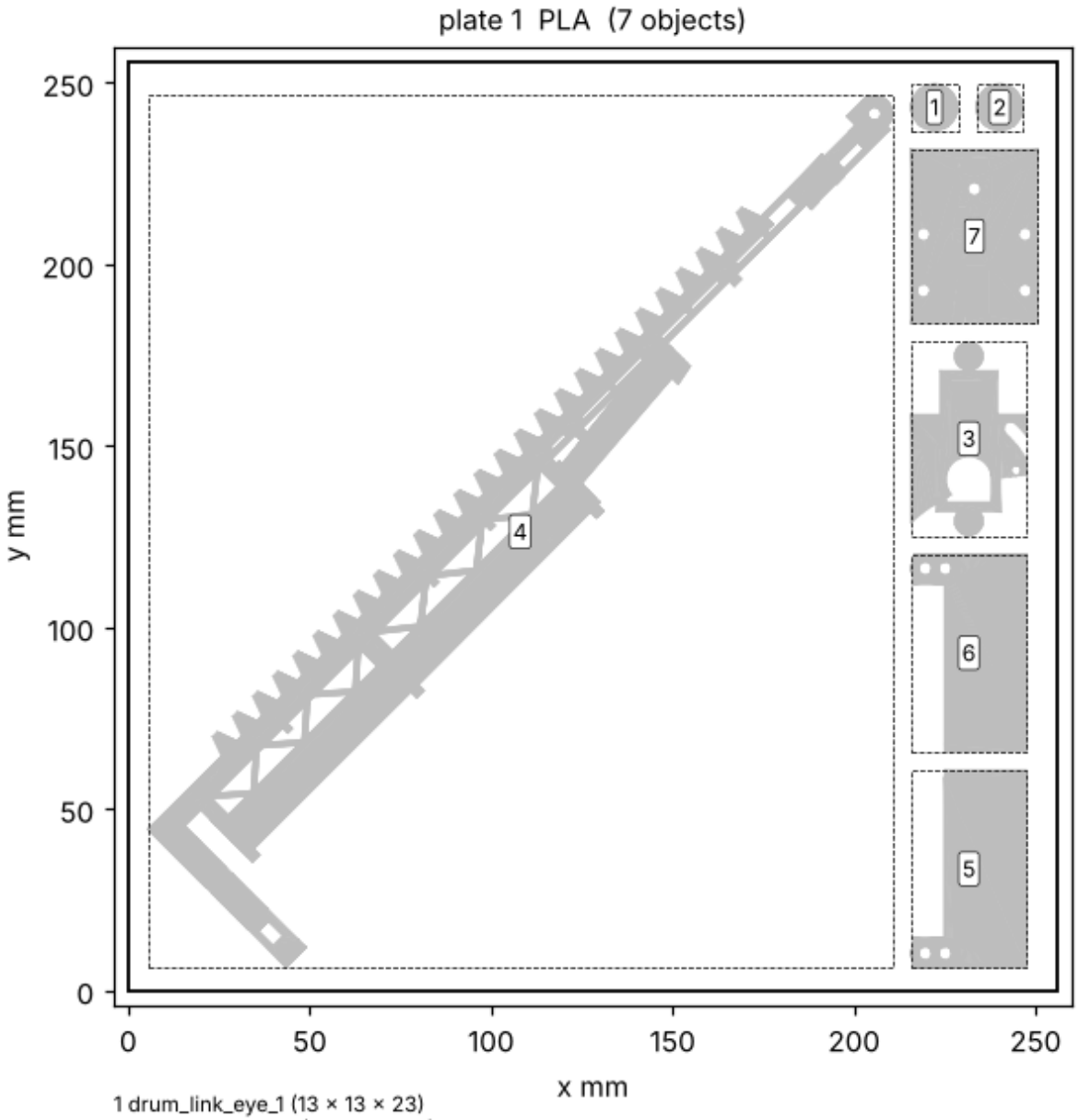
- 4 walls, 5 top and 5 bottom layers
- 40 % gyroid infill
- Nozzle 220 C, bed 55 to 60 C, textured PEI plate
- Door open or top vented

PPA-CF

- Hardened steel nozzle (0.4 or 0.6 mm), never a brass nozzle
- 0.2 mm layers
- 6 walls, 6 top and 6 bottom layers
- 60 % gyroid infill
- Dry the filament first (80 C for 8 h or more) and print from a dry box
- Enclosure closed, chamber heated if the printer can
- Nozzle 290 to 320 C, bed 100 to 110 C (check the spool label)

plate_1_PLA.3mf

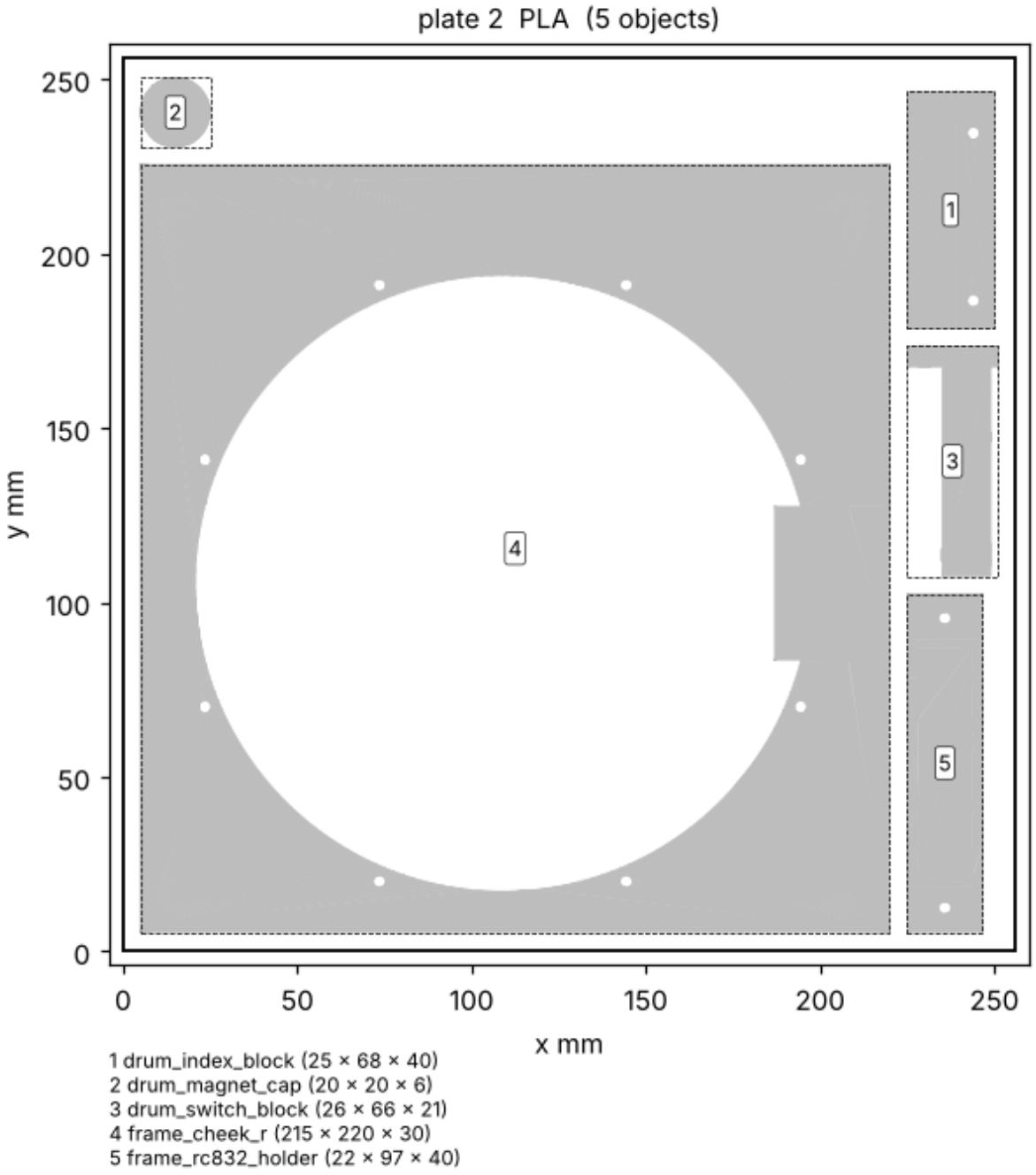
PLA, 7 objects, about 95 g and 2.6 h.



1 drum_link_eye_1 (13 × 13 × 23)
2 drum_link_eye_2 (13 × 13 × 23)
3 drum_n20_cradle (32 × 54 × 20)
4 launch_carriage (205 × 240 × 43)
5 launch_guard_l_c (32 × 54 × 26)
6 launch_guard_r_c (32 × 54 × 26)
7 sensors_sw_rail_low (35 × 48 × 12)

plate_2_PLA.3mf

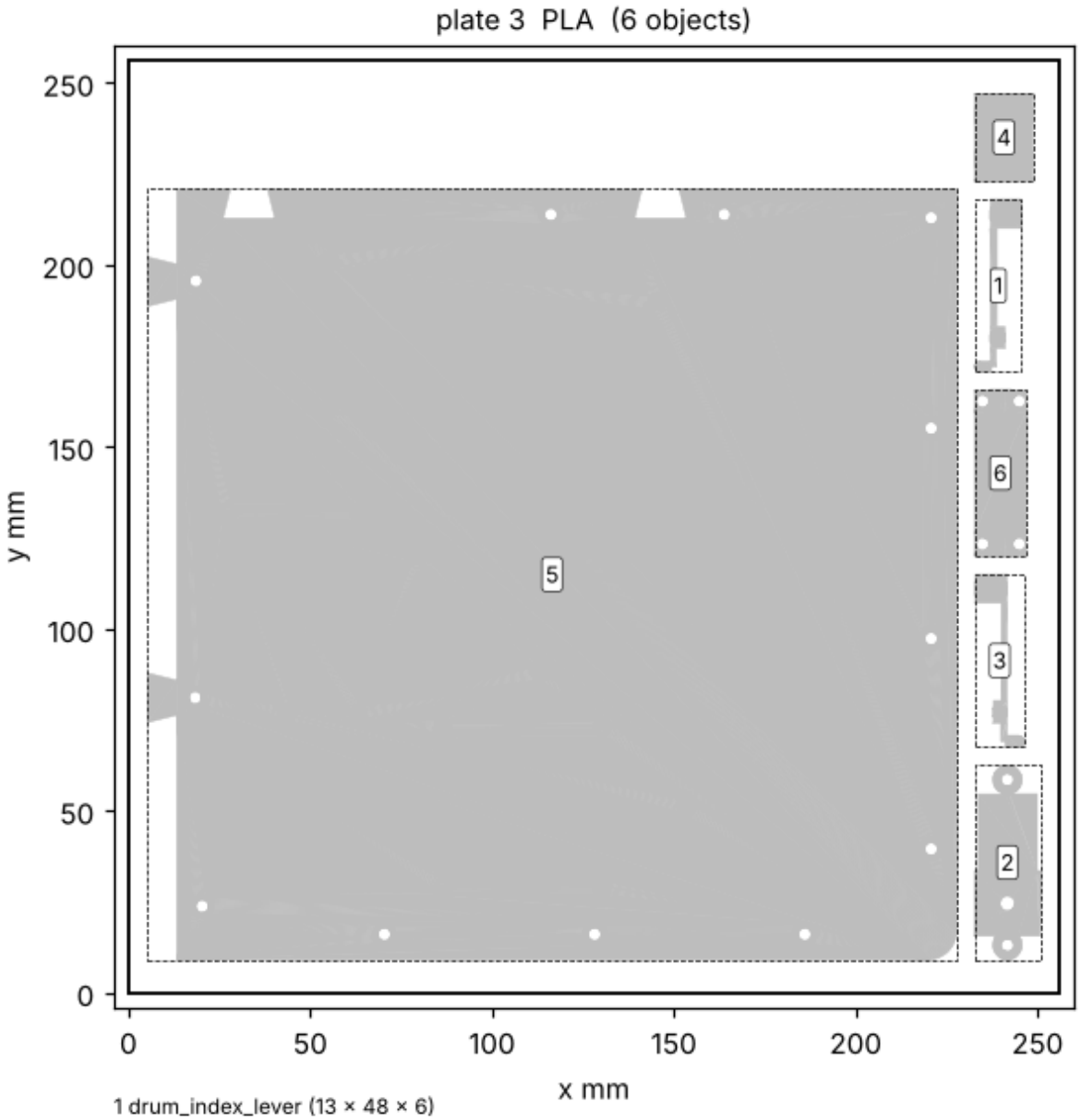
PLA, 5 objects, about 281 g and 7.3 h.



1 drum_index_block (25 × 68 × 40)
2 drum_magnet_cap (20 × 20 × 6)
3 drum_switch_block (26 × 66 × 21)
4 frame_cheek_r (215 × 220 × 30)
5 frame_rc832_holder (22 × 97 × 40)

plate_3_PLA.3mf

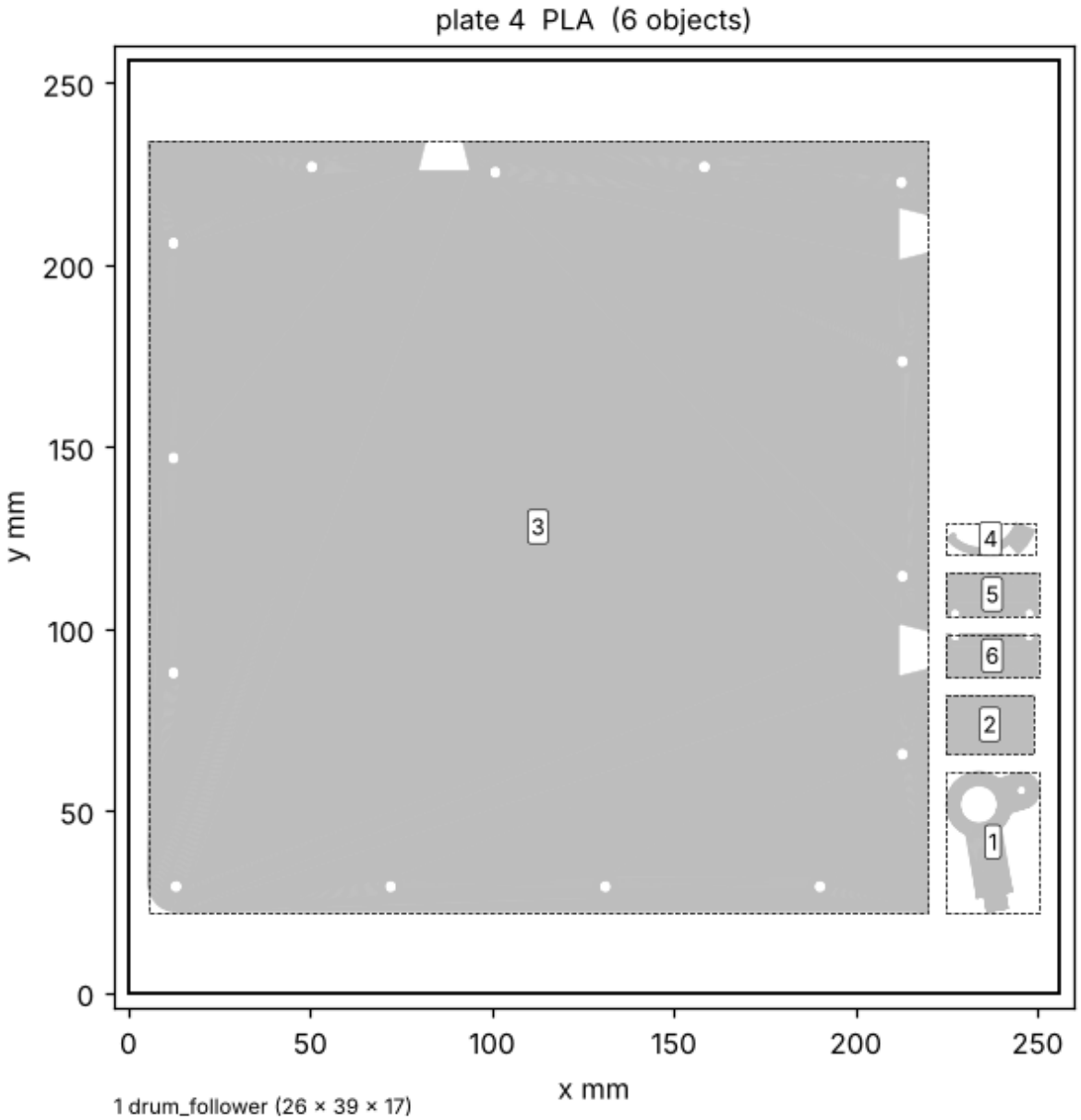
PLA, 6 objects, about 318 g and 8.1 h.



1 drum_index_lever (13 × 48 × 6)
2 drum_n20_strap (18 × 54 × 3)
3 drum_sw_lever_2 (14 × 48 × 6)
4 frame_cable_anchor_1 (16 × 24 × 10)
5 frame_deck_2 (222 × 212 × 8)
6 launch_aim_cap (14 × 46 × 19)

plate_4_PLA.3mf

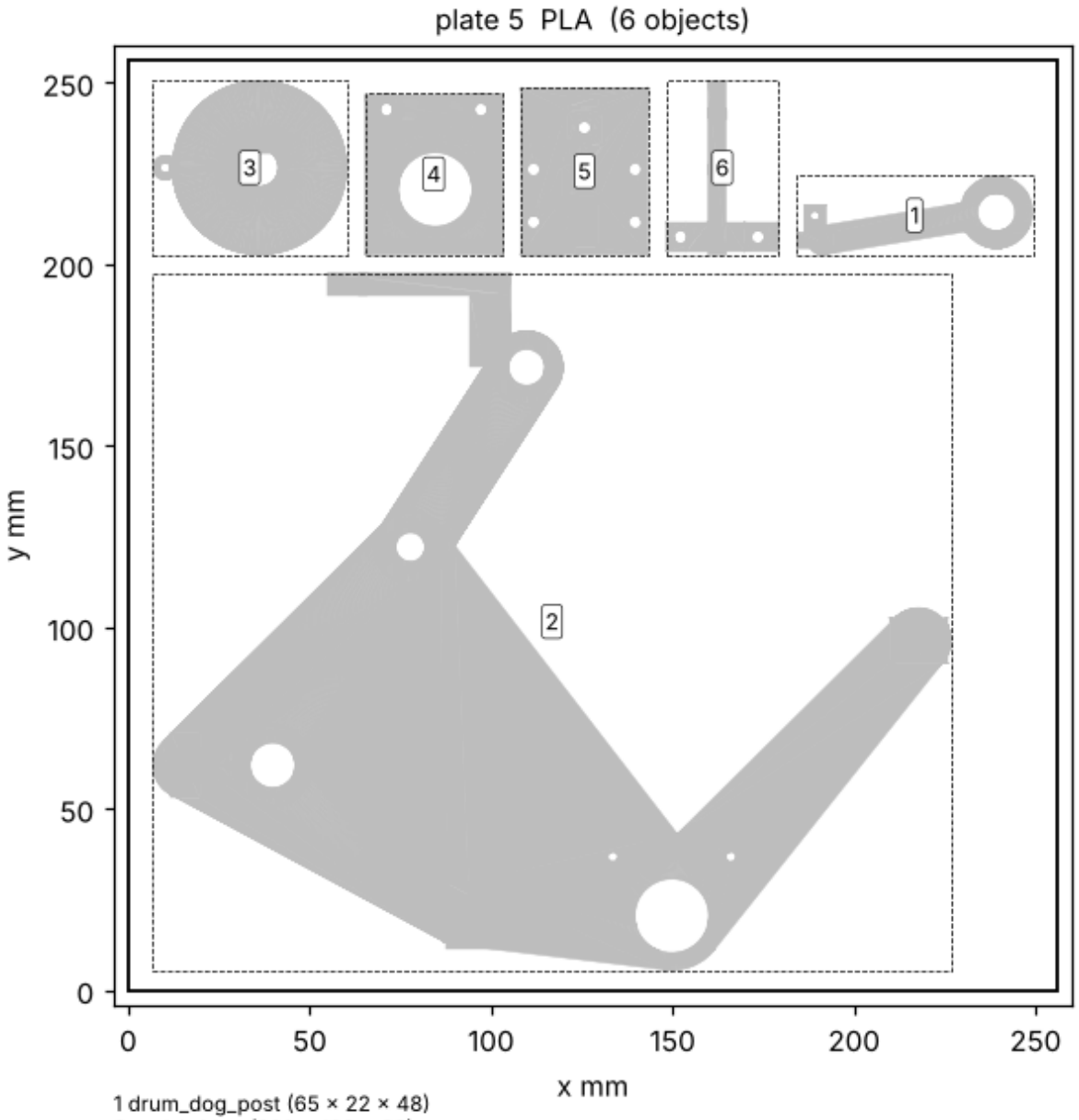
PLA, 6 objects, about 314 g and 7.9 h.



1 drum_follower (26 × 39 × 17)
2 frame_cable_anchor_2 (24 × 16 × 10)
3 frame_deck_4 (214 × 212 × 8)
4 launch_aim_drag_pad (24 × 9 × 4)
5 sensors_as5600_strip_bot (25 × 12 × 3)
6 sensors_as5600_strip_top (25 × 12 × 3)

plate_5_PLA.3mf

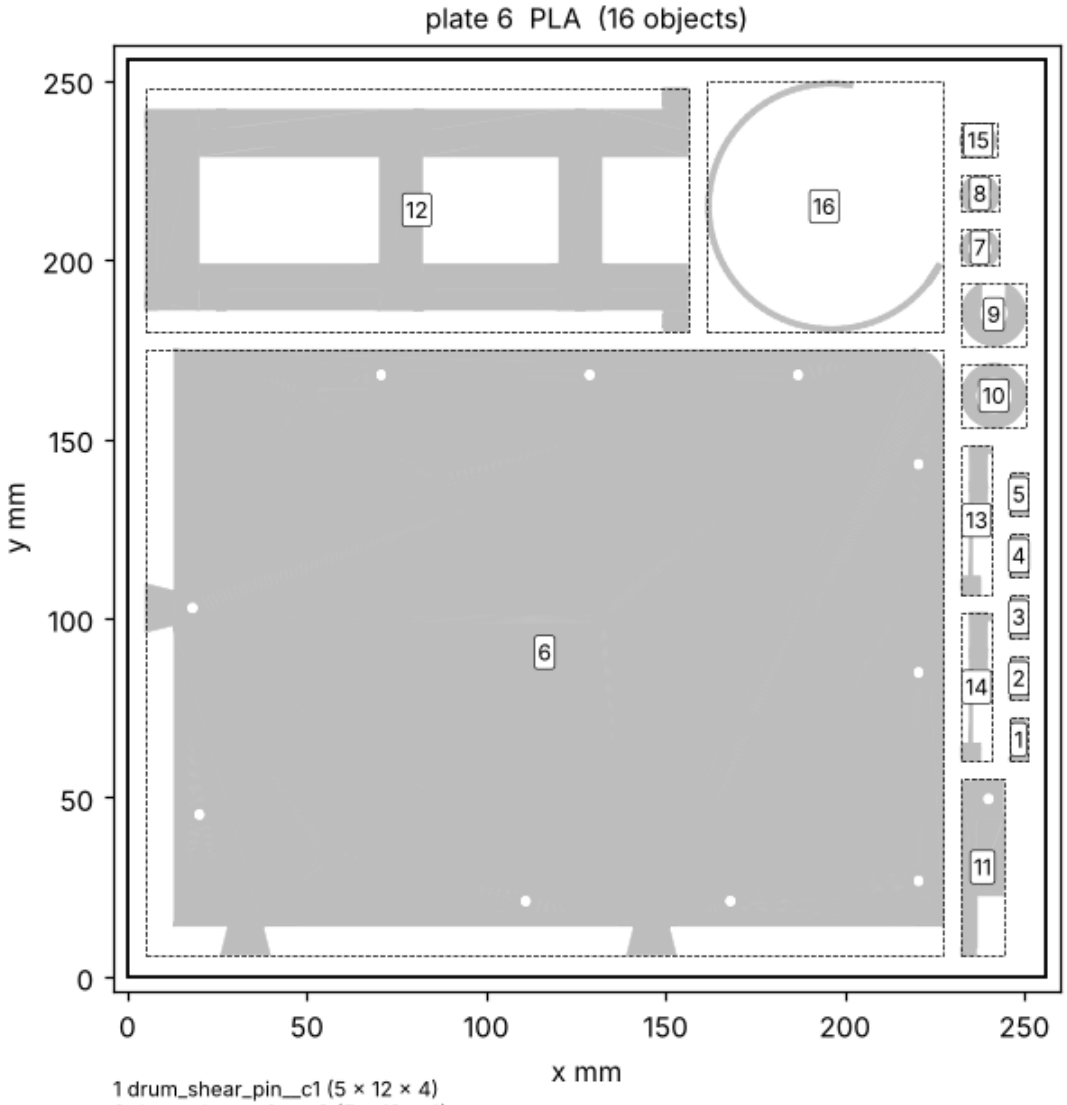
PLA, 6 objects, about 174 g and 4.9 h.



1 drum_dog_post (65 × 22 × 48)
2 reel_bridge (220 × 192 × 88)
3 reel_cup (53 × 48 × 18)
4 reel_head_p1b (38 × 44 × 16)
5 reel_sw_base (35 × 46 × 12)
6 sensors_sw_rail_low_bracket (31 × 48 × 74)

plate_6_PLA.3mf

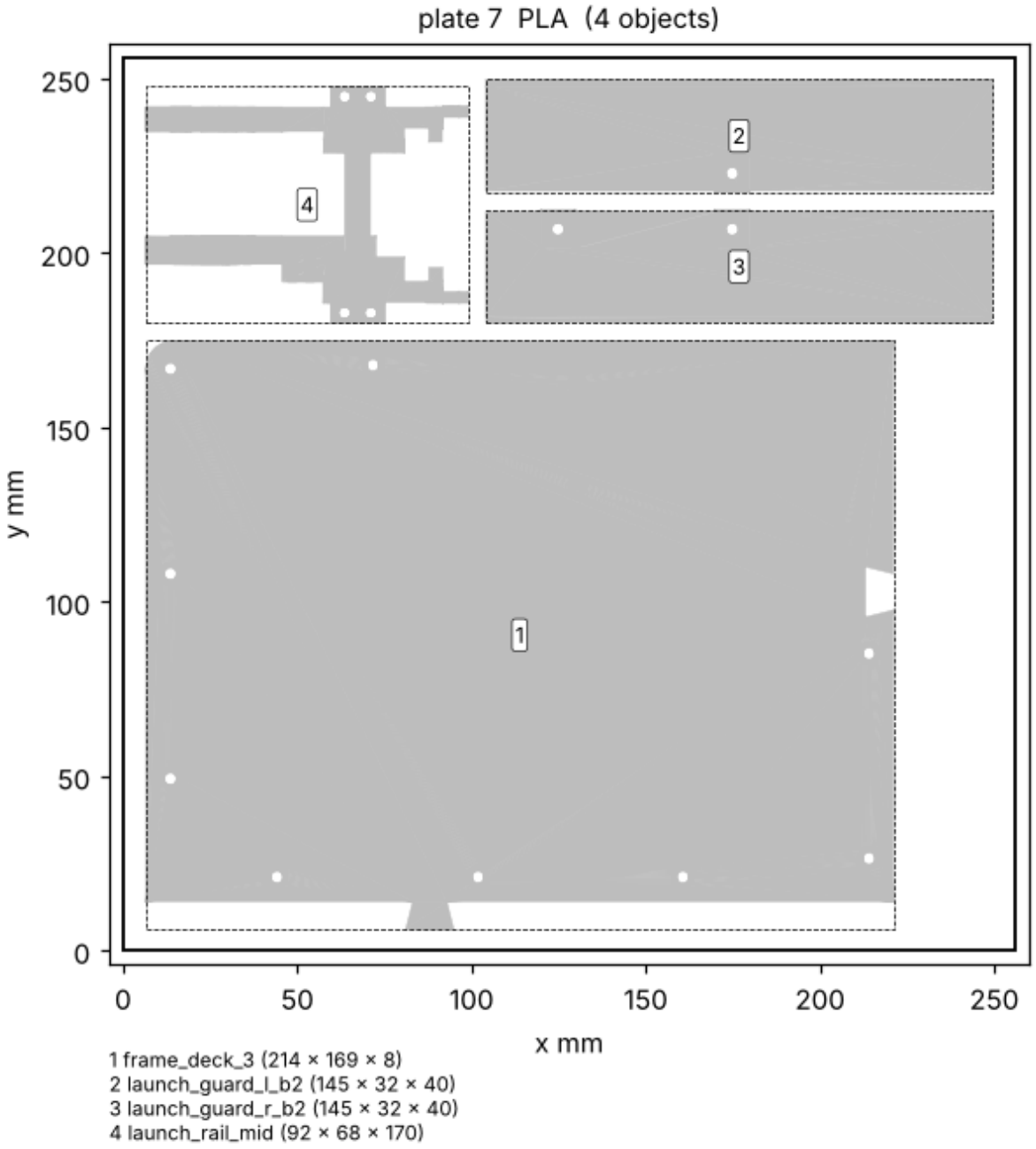
PLA, 16 objects, about 365 g and 9.4 h.



1 drum_shear_pin__c1 (5 × 12 × 4)
2 drum_shear_pin__c2 (5 × 12 × 4)
3 drum_shear_pin__c3 (5 × 12 × 4)
4 drum_shear_pin__c4 (5 × 12 × 4)
5 drum_shear_pin__c5 (5 × 12 × 4)
6 frame_deck_1 (222 × 169 × 8)
7 launch_aim_crankpin_clip (11 × 10 × 1)
8 launch_aim_lugpin_clip (11 × 10 × 1)
9 launch_gear_clip (18 × 17 × 1)
10 launch_gear_washer (18 × 18 × 2)
11 launch_guard_r_b1 (12 × 49 × 40)
12 launch_rail_rear (151 × 68 × 40)
13 launch_sw_home (8 × 42 × 6)
14 launch_sw_seated (8 × 42 × 6)
15 reel_cup_clip (10 × 10 × 2)
16 reel_shroud (66 × 70 × 30)

plate_7_PLA.3mf

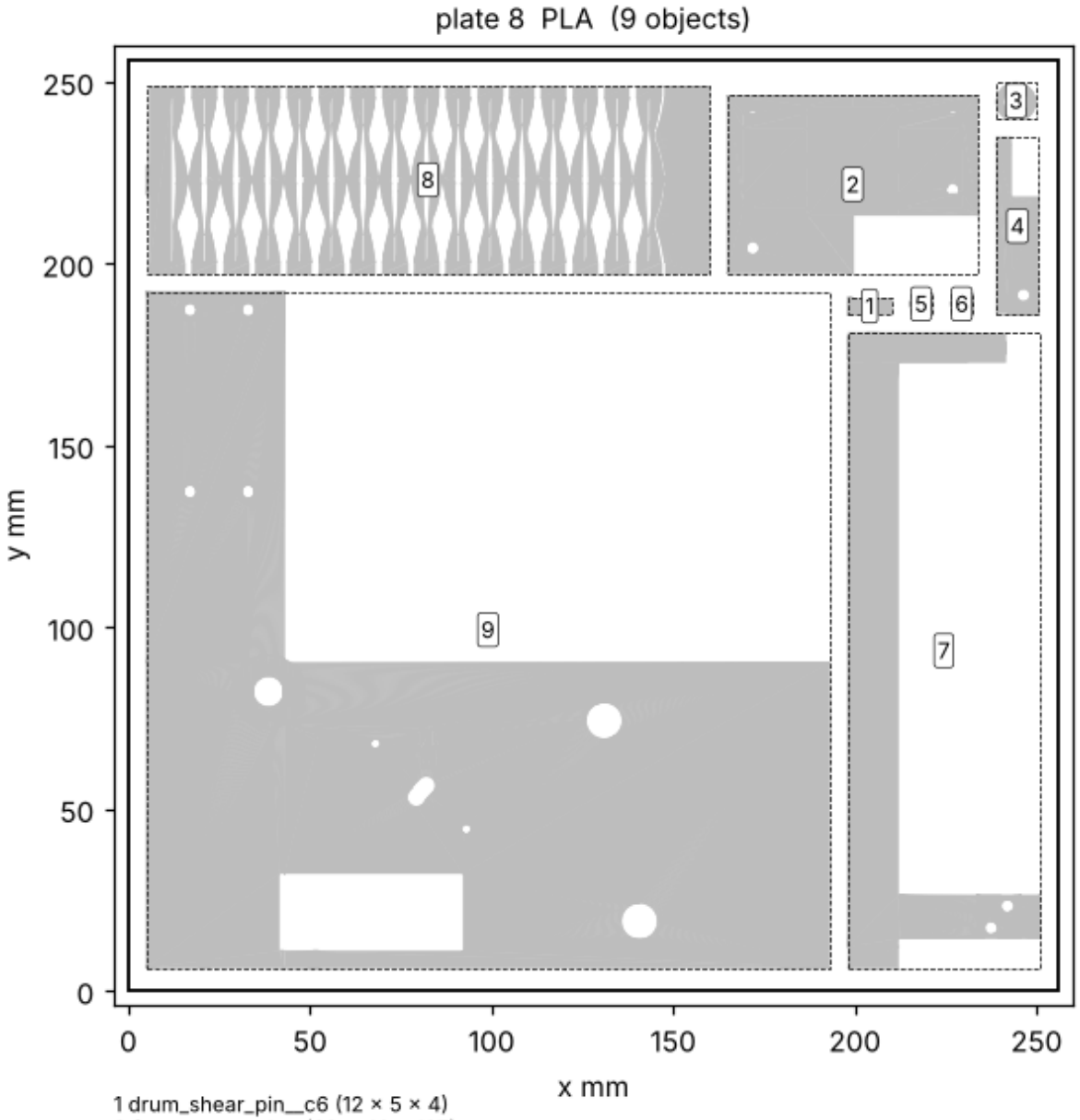
PLA, 4 objects, about 450 g and 12.4 h.



1 frame_deck_3 (214 × 169 × 8)
2 launch_guard_l_b2 (145 × 32 × 40)
3 launch_guard_r_b2 (145 × 32 × 40)
4 launch_rail_mid (92 × 68 × 170)

plate_8_PLA.3mf

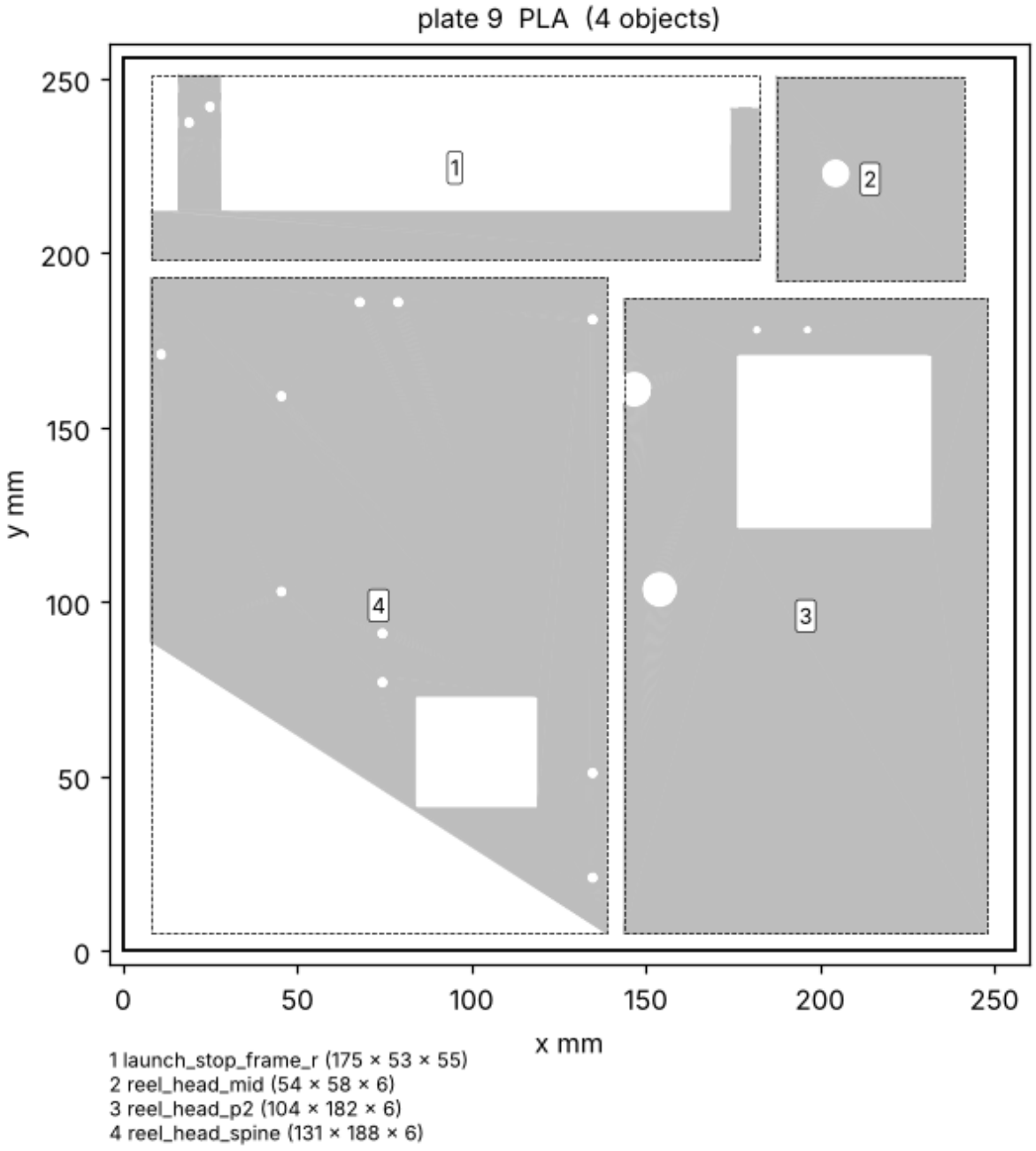
PLA, 9 objects, about 345 g and 9.9 h.



1 drum_shear_pin__c6 (12 × 5 × 4)
2 frame_motor_leg (69 × 49 × 187)
3 launch_aim_idler_clip (11 × 10 × 1)
4 launch_guard_l_b1 (12 × 49 × 40)
5 launch_muzzle_washer_l (6 × 6 × 1)
6 launch_muzzle_washer_r (6 × 6 × 1)
7 launch_stop_frame_l (53 × 175 × 55)
8 launch_stop_spring_l (155 × 52 × 14)
9 reel_head_front (188 × 186 × 6)

plate_9_PLA.3mf

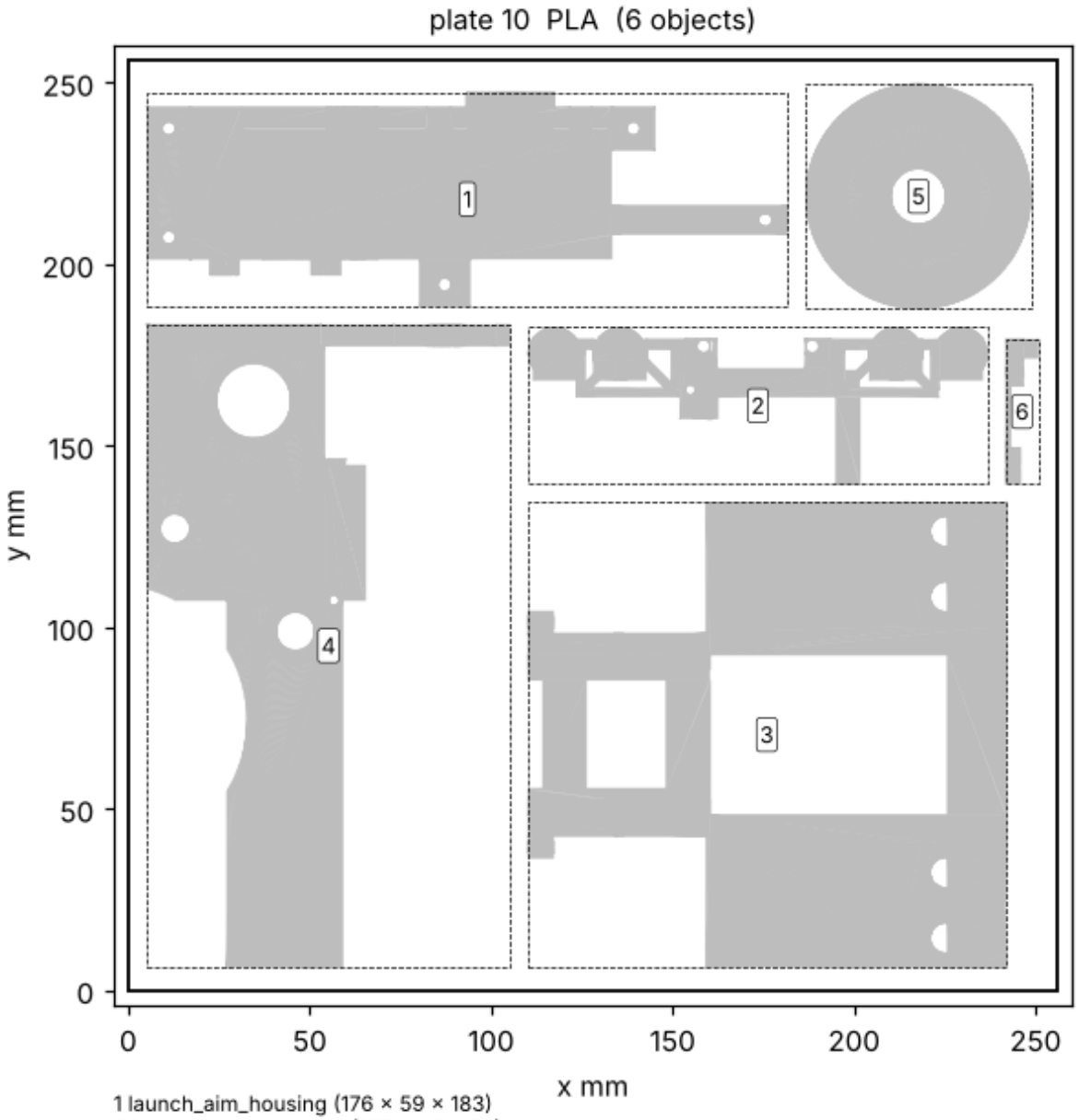
PLA, 4 objects, about 250 g and 6.6 h.



1 launch_stop_frame_r (175 × 53 × 55)
2 reel_head_mid (54 × 58 × 6)
3 reel_head_p2 (104 × 182 × 6)
4 reel_head_spine (131 × 188 × 6)

plate_10_PLA.3mf

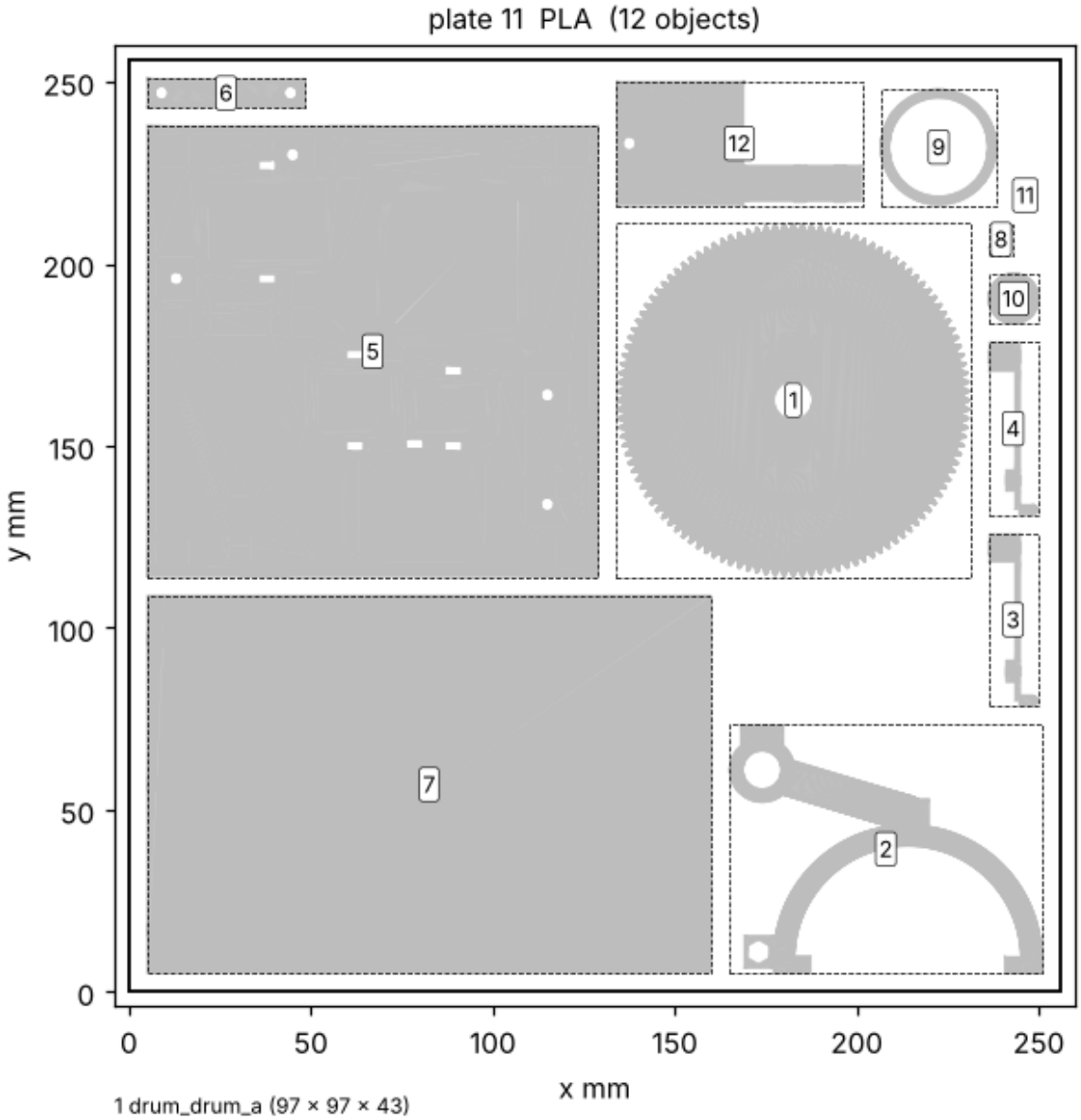
PLA, 6 objects, about 434 g and 12.1 h.



1 launch_aim_housing (176 × 59 × 183)
2 launch_carriage_yoke (127 × 43 × 28)
3 launch_rail_front (132 × 128 × 48)
4 reel_head_p1 (100 × 177 × 37)
5 reel_spool (62 × 62 × 22)
6 sensors_sw_gap_lever (9 × 40 × 8)

plate_11_PLA.3mf

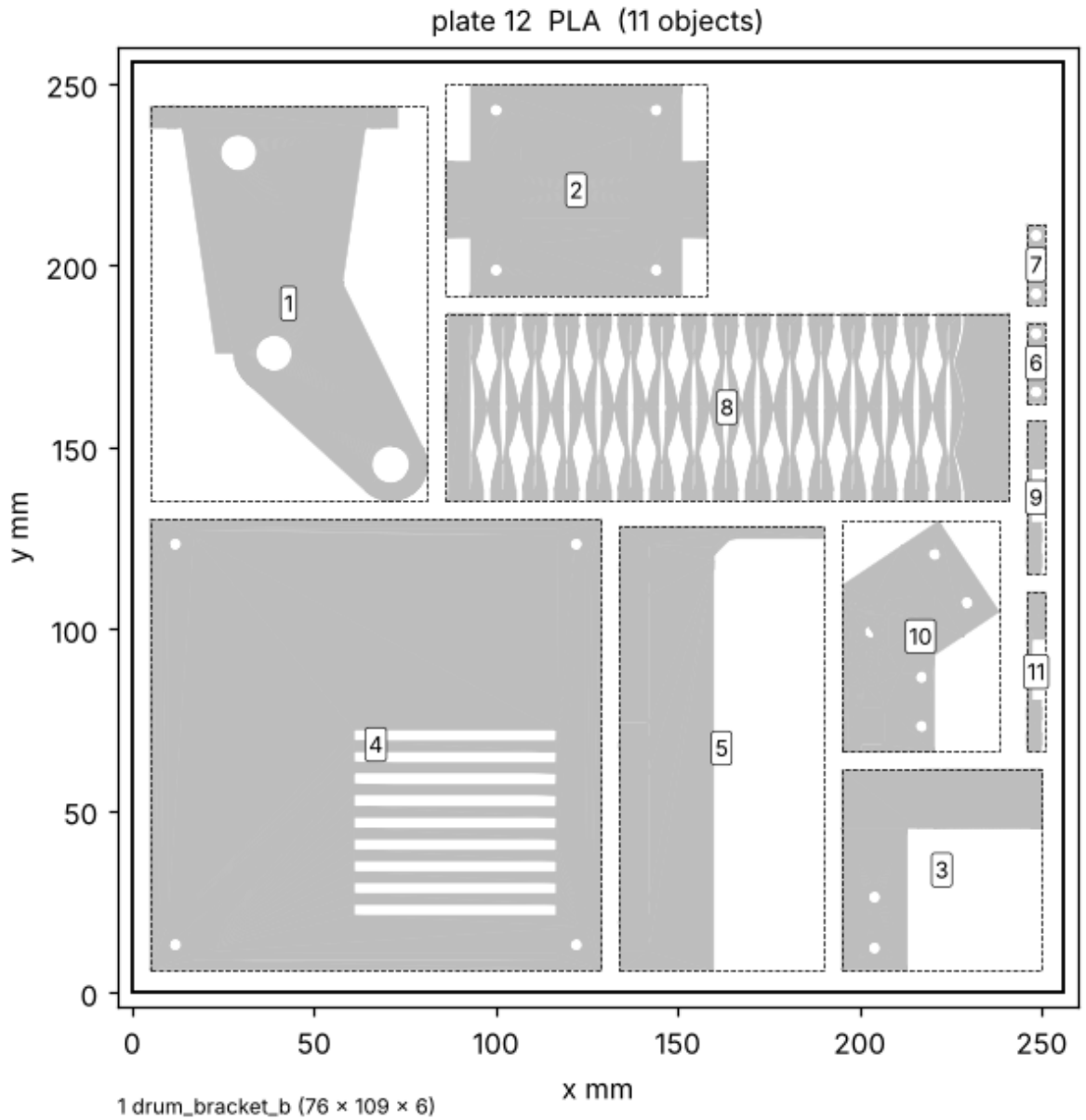
PLA, 12 objects, about 372 g and 9.7 h.



1 drum_drum_a (97 × 97 × 43)
2 drum_fork (86 × 68 × 51)
3 drum_sw_lever_0 (14 × 48 × 6)
4 drum_sw_lever_1 (14 × 48 × 6)
5 frame_ebay (124 × 124 × 58)
6 frame_ebay_esp_clamp (43 × 8 × 4)
7 frame_guard_front (155 × 104 × 16)
8 launch_aim_drag_shim (7 × 9 × 4)
9 launch_drive_washer (32 × 32 × 1)
10 launch_muzzle_roller (14 × 14 × 5)
11 launch_seat_clip (6 × 6 × 2)
12 sensors_as5600_carrier (68 × 34 × 11)

plate_12_PLA.3mf

PLA, 11 objects, about 401 g and 11.6 h.



1 drum_bracket_b (76 × 109 × 6)
2 frame_camera_mount (72 × 58 × 236)
3 frame_can_stay (55 × 55 × 71)
4 frame_ebay_lid (124 × 124 × 7)
5 frame_guard_hood (56 × 122 × 14)
6 launch_rail_splice_l (5 × 22 × 3)
7 launch_rail_splice_r (5 × 22 × 3)
8 launch_stop_spring_r (155 × 52 × 14)
9 reel_sw_lever (5 × 42 × 8)
10 sensors_sw_gap (43 × 63 × 12)
11 sensors_sw_rail_low_lever (5 × 44 × 8)

plate_13_PLA.3mf

PLA, 1 objects, about 82 g and 2.3 h.

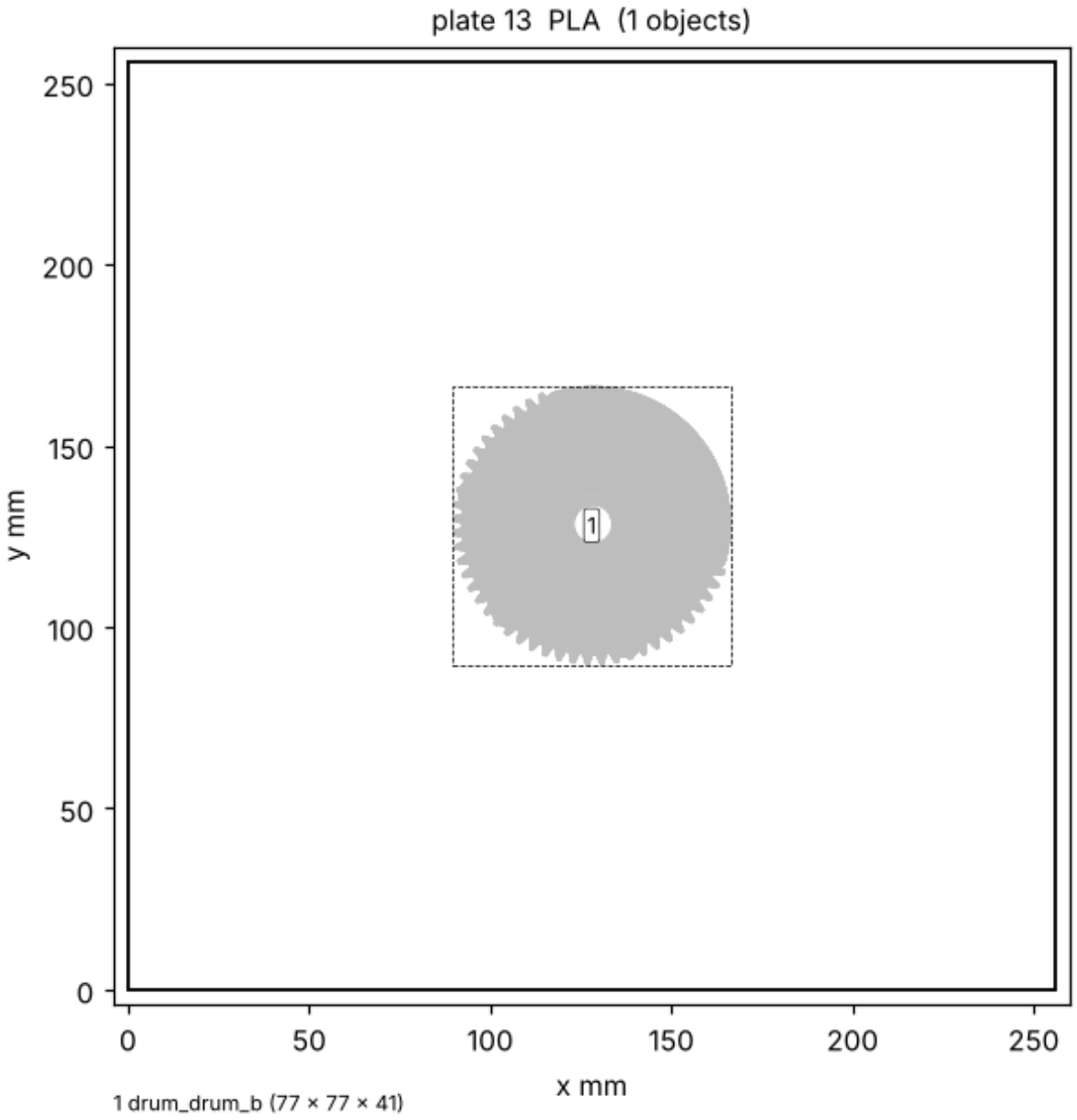


plate_14_PLA.3mf

PLA, 1 objects, about 65 g and 1.9 h.

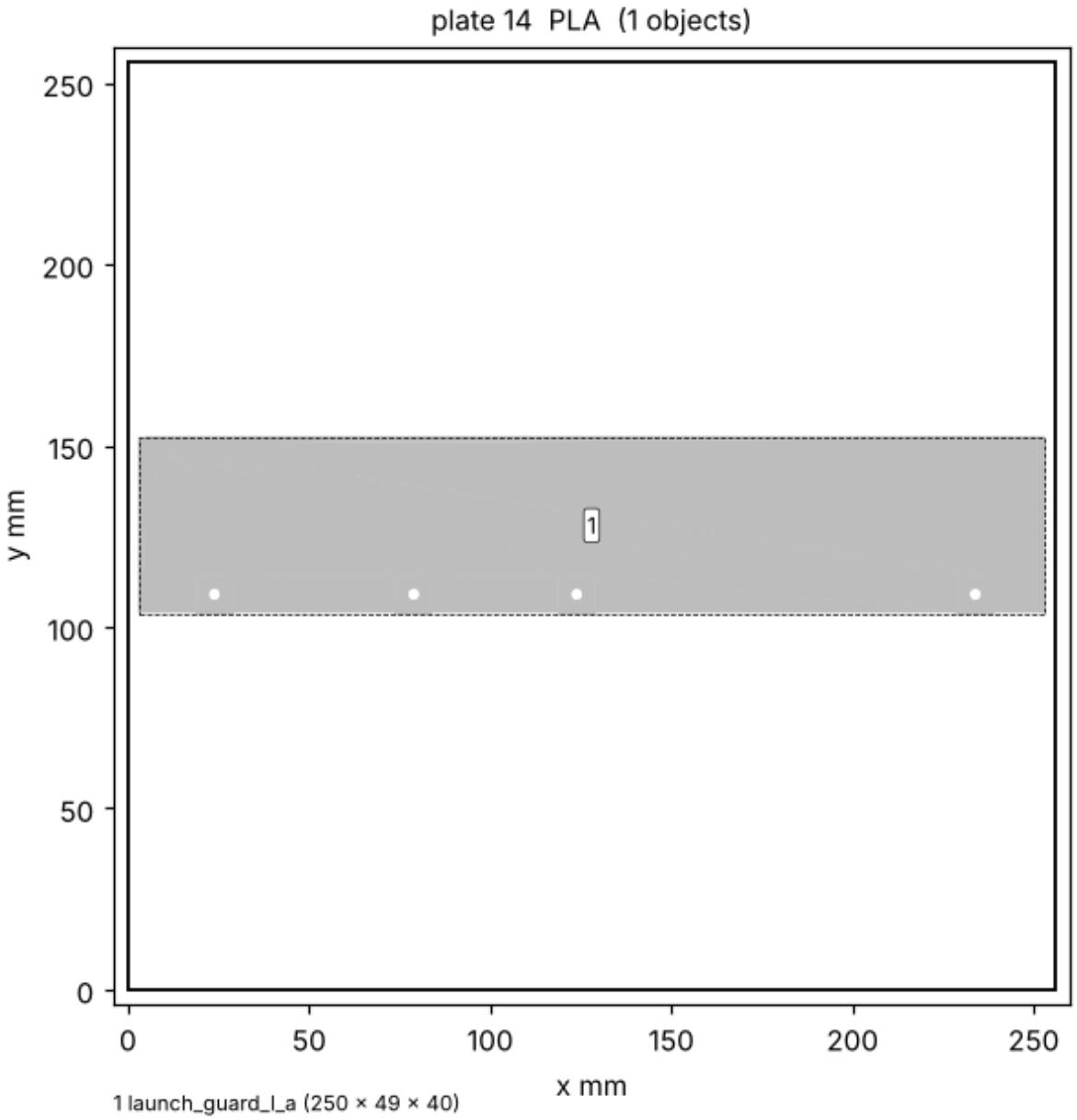


plate_15_PLA.3mf

PLA, 1 objects, about 65 g and 1.9 h.

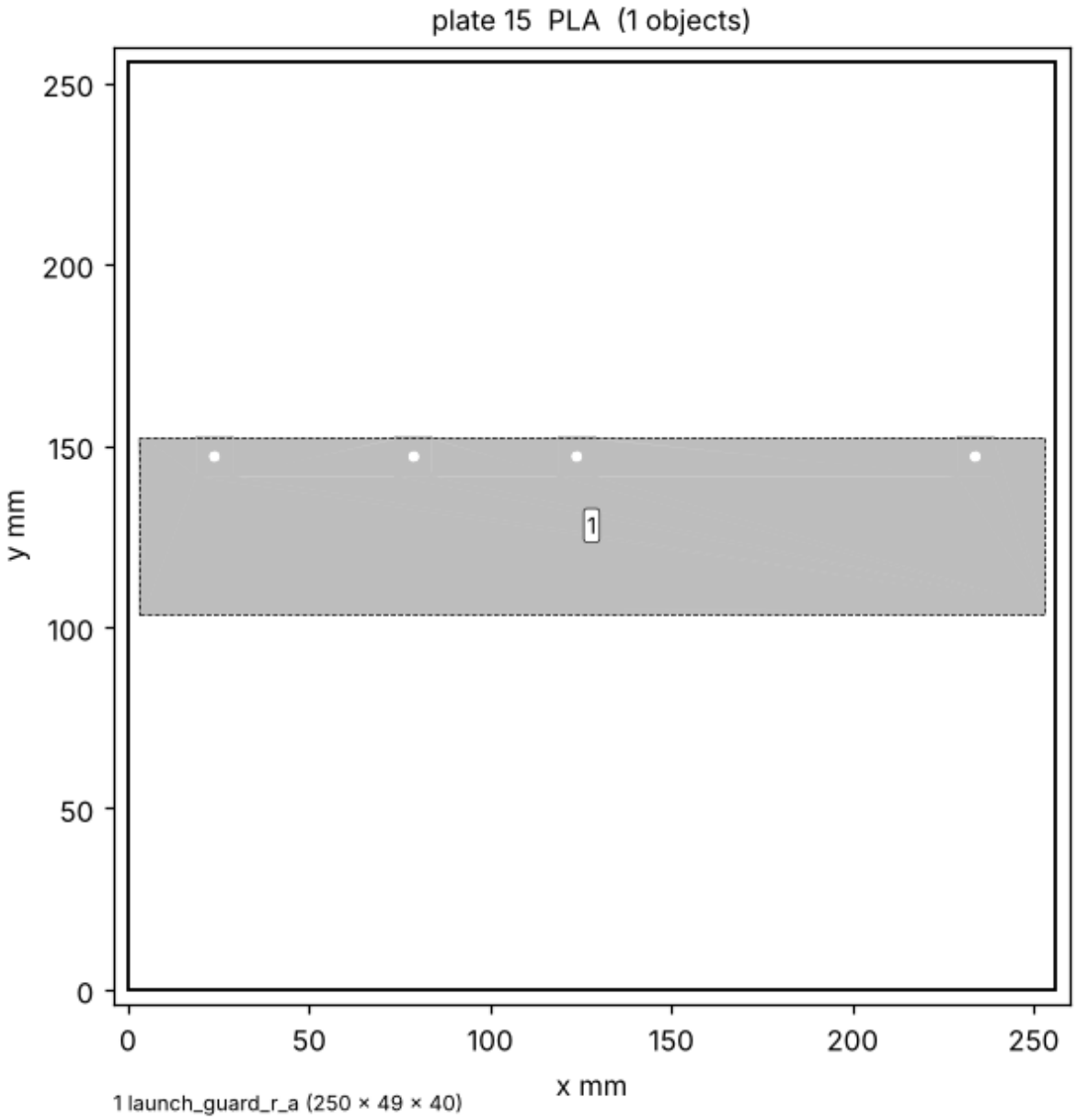
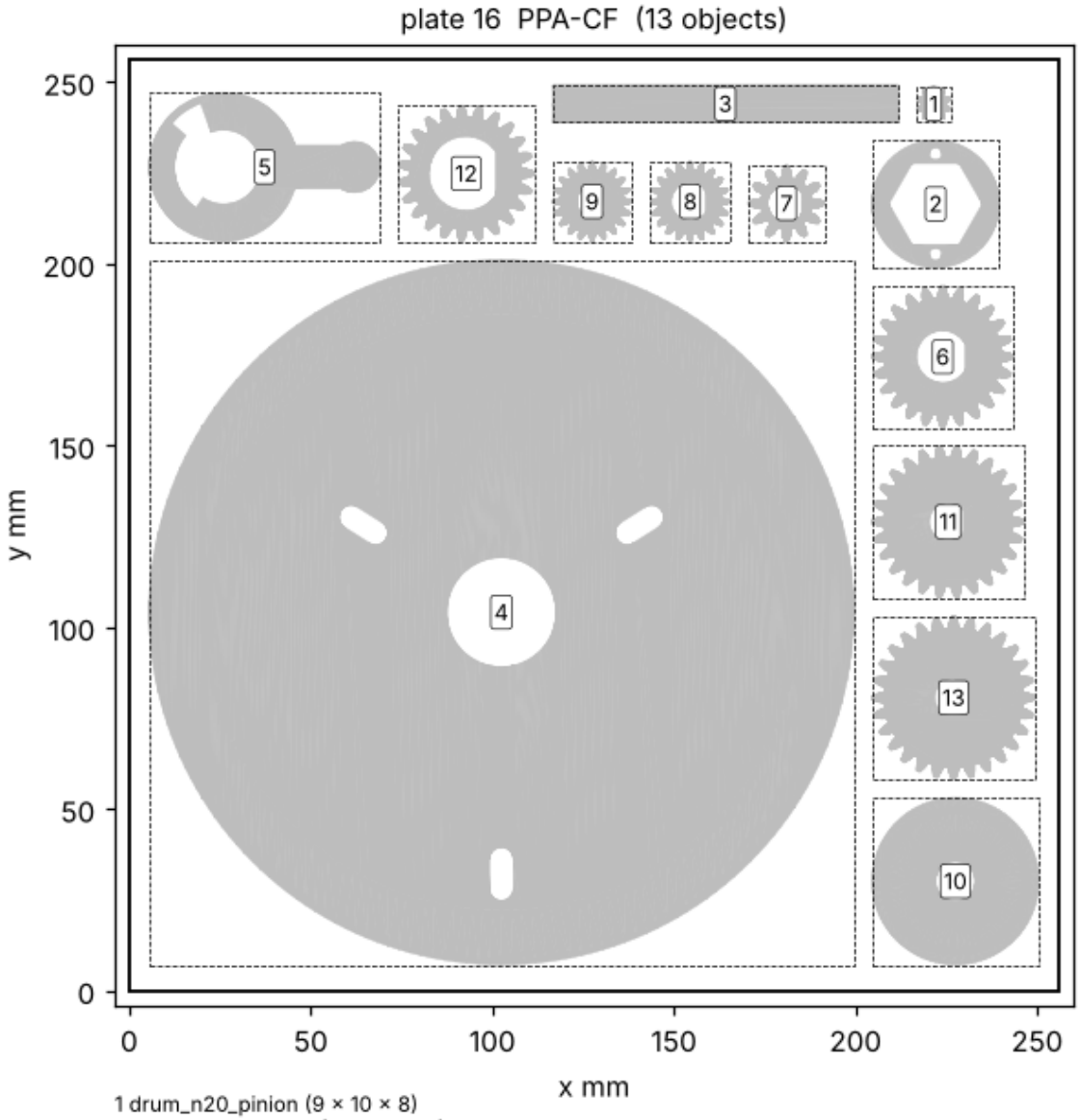


plate_16_PPA-CF.3mf

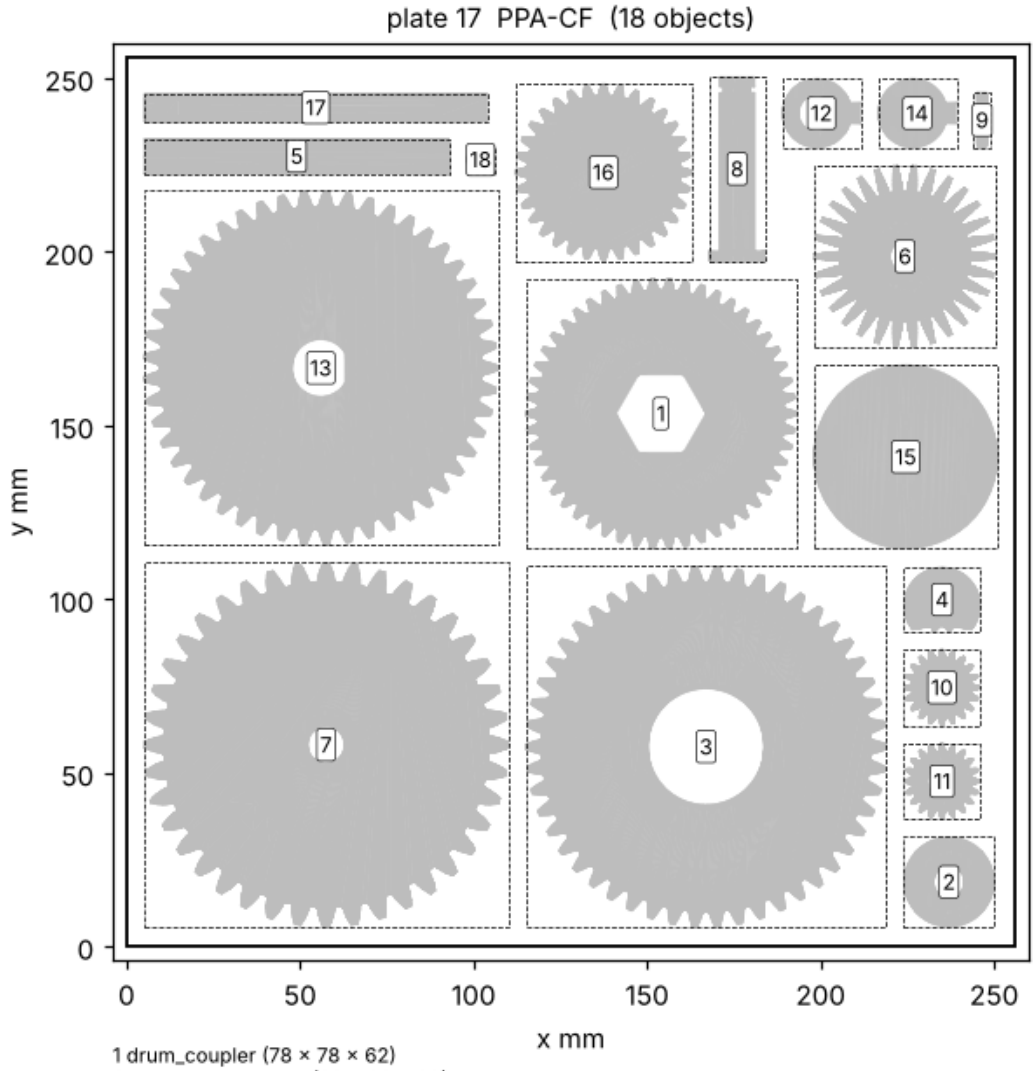
PPA-CF, 13 objects, about 401 g and 18.1 h.



1 drum_n20_pinion (9 × 10 × 8)
2 drum_reel_washer (35 × 35 × 1)
3 drum_rod (95 × 10 × 9)
4 frame_motor_adapter (194 × 194 × 16)
5 launch_aim_arm (64 × 41 × 19)
6 reel_b1g (39 × 39 × 6)
7 reel_b1p (21 × 21 × 11)
8 reel_b2g (22 × 22 × 7)
9 reel_b3a (22 × 22 × 8)
10 reel_drum (46 × 46 × 43)
11 reel_ga (42 × 42 × 7)
12 reel_gc (37 × 37 × 7)
13 reel_gk (45 × 45 × 12)

plate_17_PPA-CF.3mf

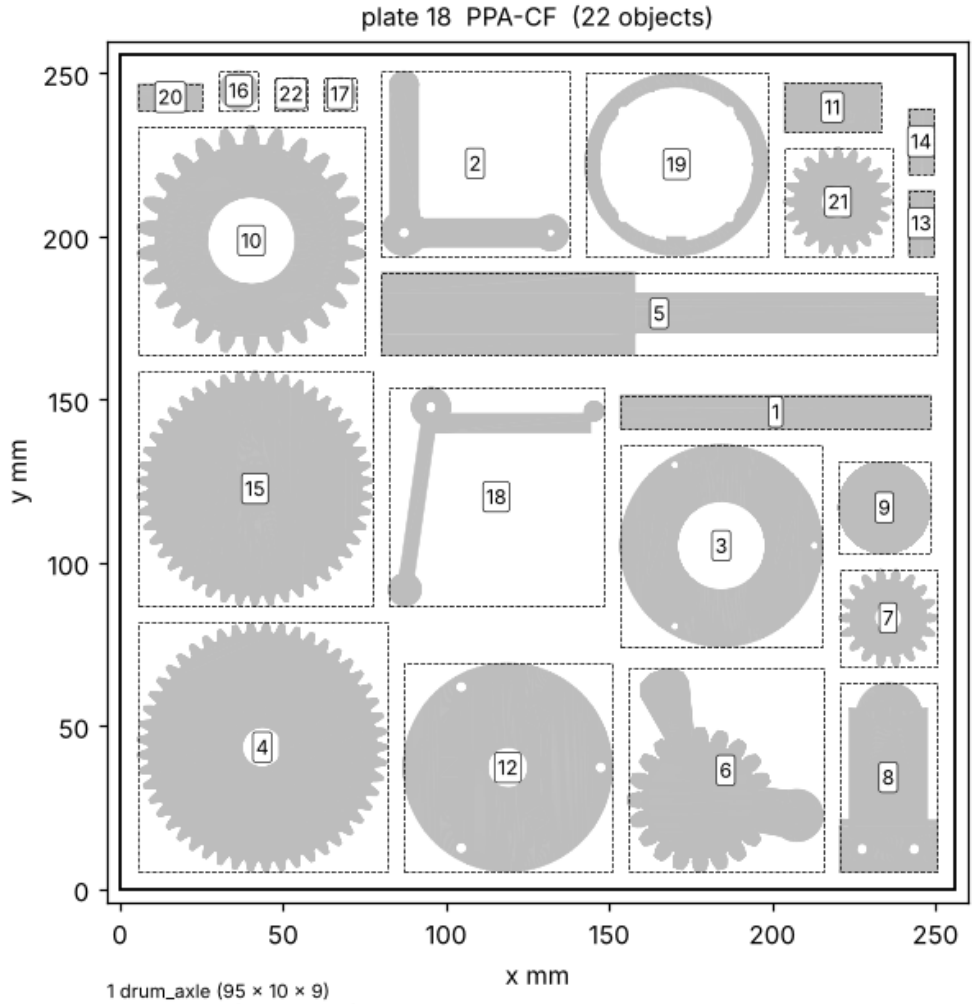
PPA-CF, 18 objects, about 569 g and 26.0 h.



1 drum_coupler (78 × 78 × 62)
2 drum_coupler_hub (26 × 26 × 32)
3 drum_gear_reel (104 × 104 × 19)
4 drum_idler (22 × 19 × 54)
5 drum_shift_rail (88 × 10 × 9)
6 launch_aim_bevel30 (53 × 52 × 10)
7 launch_big_gear (105 × 105 × 27)
8 launch_gear_pin (16 × 53 × 12)
9 launch_seat_pushrod (5 × 16 × 4)
10 reel_b2p (22 × 22 × 8)
11 reel_b3b (22 × 22 × 9)
12 reel_c (23 × 20 × 58)
13 reel_c1 (102 × 102 × 12)
14 reel_cs_c (23 × 20 × 40)
15 reel_d (53 × 53 × 3)
16 reel_gb (51 × 51 × 7)
17 reel_s (99 × 8 × 7)
18 reel_ut (8 × 8 × 104)

plate_18_PPA-CF.3mf

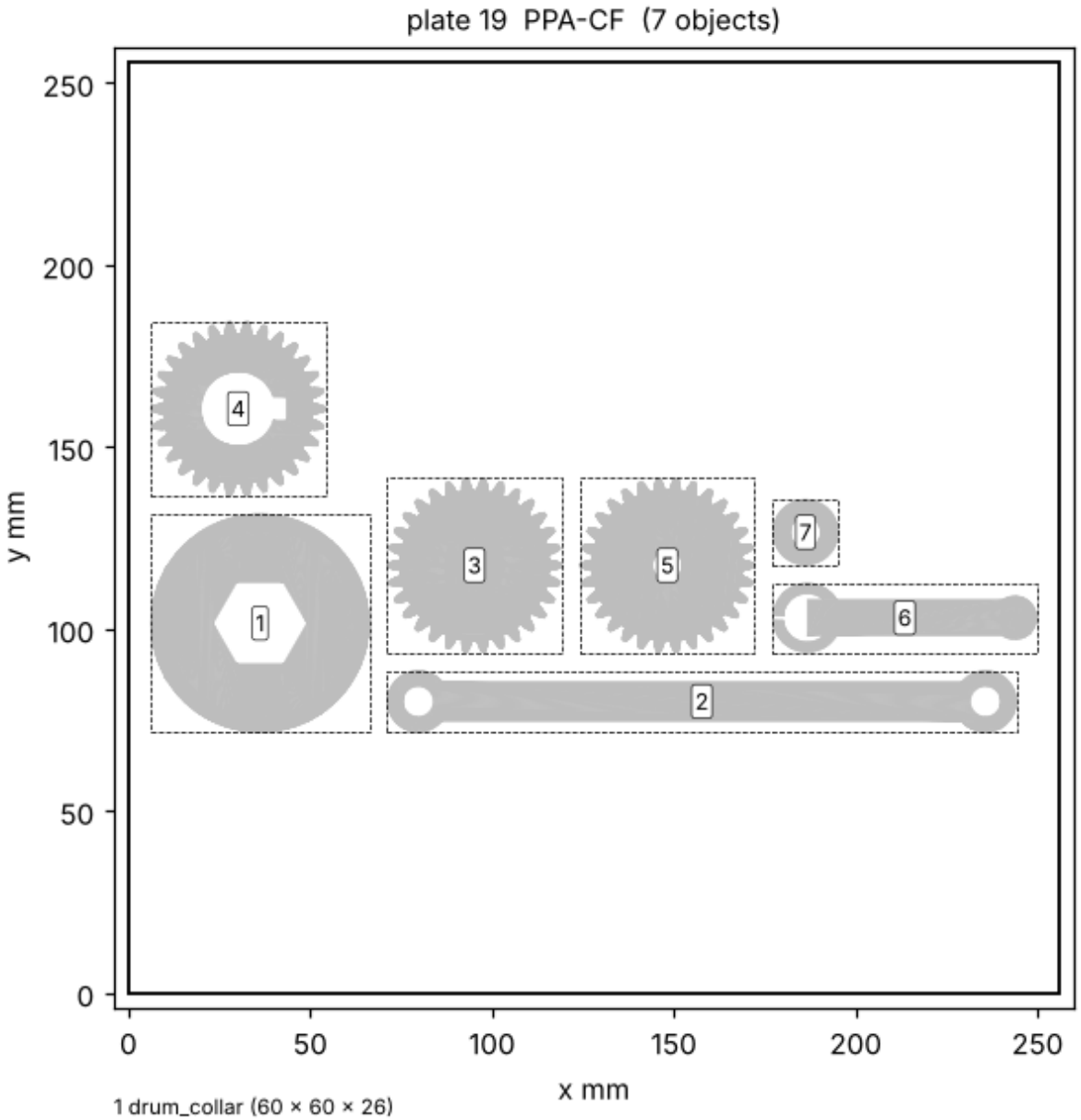
PPA-CF, 22 objects, about 369 g and 18.1 h.



1 drum_axle (95 × 10 × 9)
2 drum_crank (58 × 56 × 10)
3 drum_dogring_launch (62 × 62 × 10)
4 drum_idler_gear (76 × 76 × 7)
5 drum_shaft (170 × 25 × 22)
6 launch_aim_crank (60 × 62 × 38)
7 launch_aim_idler (30 × 30 × 12)
8 launch_aim_lug (30 × 58 × 28)
9 launch_aim_worm (28 × 28 × 54)
10 launch_drive_gear (70 × 70 × 29)
11 launch_seat_plunger (30 × 15 × 20)
12 launch_sleeve (64 × 64 × 85)
13 launch_yoke_peg_l (8 × 20 × 7)
14 launch_yoke_peg_r (8 × 20 × 7)
15 reel_c1p (72 × 72 × 10)
16 reel_cs_t (12 × 12 × 24)
17 reel_k (10 × 10 × 109)
18 reel_lever (66 × 67 × 8)
19 reel_sprag_hub (56 × 56 × 14)
20 reel_stub (20 × 8 × 7)
21 reel_t1 (33 × 33 × 6)
22 reel_uc (10 × 10 × 244)

plate_19_PPA-CF.3mf

PPA-CF, 7 objects, about 118 g and 5.7 h.



1 drum_collar (60 × 60 × 26)
2 launch_aim_conrod (173 × 17 × 6)
3 launch_aim_output (48 × 48 × 60)
4 launch_aim_sun (48 × 48 × 12)
5 reel_gi (48 × 48 × 6)
6 reel_swing_arm (73 × 19 × 10)
7 reel_t (18 × 18 × 6)